

Problem statement (Page I of IV)

Company: Diamond
Year: 2008
Round: I
Case Name: IT Budget

Problem Statement - Narrative

Your client is a large financial services firm, like Citigroup or Chase. They have been approached by IBM to outsource all of their IT needs to IBM. IBM claims that it can save the firm 10% off their current annual IT budget. The current budget is 1.5B. The CIO (the client) wants to know whether to use IBM or not.

Objectives: How can IBM possibly save this firm 10% off their current IT budget?

Guidance for the interviewer

Information to be provided upon request

- There are 100k employees at the firm. Of them, 6k are in IT
- Of the 6k IT employees, 5k are internal and 1k are external
- An “external” employee is a consultant (not an actual employee of the firm), internal employees are just regular employees

Problem statement (Page II of IV)

Guidance for the interviewer

Information to be provided upon request

The Interviewee should be able to define these categories. Do not show them this table.

Area	Budget
Hardware (servers, wiring, accessories, etc.)	100M
Software	100M
Data Center (the place where the servers and hardware actually live)	100M
Network (connectivity, so that our stuff is online)	100M
Workstations, laptops, printers (things we use in the office and on the road)	100M
Other (assume fixed costs)	100M
Labor	900M * (see table below)
Total	1.5B

Problem statement (Page III of IV)

Guidance for the interviewer

Information to be provided upon request

	Salary (A)	# Employees (B)	= Total Cost
	A x B = C		
Internal Employees	\$100k per year	5,000 employees	500M per year
External Employees	\$200 per hour	1,000 employees	\$200k per hour Assume 2,000 hours per year =\$400M per year
		Total	\$900M
	Salary	# employees	Total Cost
	A x B = C		
Internal Employees	\$100k per year	5,000 employees	500M per year

Case Details (Page IV of IV)

Candidate should cover this first

1. Lead the interviewee to explain how IBM would strip off costs from the following categories, to total \$150M in savings

Candidate should cover this next (math section)

Area	Current Budget	With IBM	Savings
Hardware	100M	?	?
Software	100M	?	?
Data Center	100M	?	?
Network	100M	?	?
Workstations/Laptops	100M	?	?
Other (fixed)	100M	100M (they're fixed!)	0 (they're fixed!)
Labor (internal and external)	900M	?	?
Total	1.5B	1,350,000,000	150M

Problem statement (Page I of IV)

Company: Booz
Year: 2008
Round: II
Case Name: Customer churn

Problem Statement - Narrative

Your client is a telecom wireless service provider. Off late they have started facing a customer churn ratio of 3% which is higher than the industry average of 2%.

Customer acquisition cost for your client is \$400 per customer.

Objectives:

- 1) Find the reason for the increase in customer churn ratio.
- 2) In real life how would you go about collecting data to investigate such problems?

Guidance for the interviewer

- Totally qualitative case.
- In this case, the problem is with the indirect sales channel (small shops which sell products and services from multiple providers). There are not enough incentives for the sellers to promote the products from our client. Hence, the indirect sales people tend to promote products from other providers to our customers leading to a higher churn ratio.
- Let the interviewee come up with as many possible causes for the problem as possible. You can use the “what else” approach.
- Focus on how the interviewee structures the problem and whether he/she covers the main points. A possible solution structure is provided at the end of case.

Case Details (Page II of IV)

Candidate should cover this first:

Ask the interviewee to come up with potential sources of increase in customer churn ratio. Let him/her brain storm on the possible high level 'pain areas'.

Possible questions to guide the interviewee:

- Have you ever shopped for a cellular phone? Where did you buy it from?
- What factors do you think affect the decision making process of a cellular phone customer?

Possible answers:

- The high level 'pain areas' can either be Pricing or Services or Sales/Advertising

Candidate should cover this next:

Explore each high level 'pain area' in more details

Contd...

Possible questions to guide the interviewee:

- How do you regard the competition in the wireless telecom industry in the United States? Is it fierce, mild or non-existent? (the interviewee should correctly guess that it is a fairly competitive industry even though there are only a limited number of players)
- Given that it is a fairly competitive industry, do you think Pricing can be a issue? (the interviewee should realize at this point that Pricing could not be a 'pain area')
- How would you further sub-divide the Services bucket? (possible sub-divisions include Flexibility of plans, customer support, voice quality and coverage area, add-ons like ringtones, picture messaging, etc. Once the interviewee starts hitting the wall, you can ask him to move on to the next 'pain area')

Case Details (Page III of IV)

Candidate should cover this next:

Explore Sales 'pain area' in more details

Possible questions to guide the interviewee:

- What do you think are the major sales channels for wireless service providers? (possible answers include direct/company owned sales channels and indirect/third party sales channels)
- Which one do you think is more likely to have a problem? Why? (the interviewee should correctly identify the control problems in the indirect sales channel. He/she should question the incentive structure for this particular channel and ask questions about reward mechanism in place)
- What steps do you suggest to mitigate this problem? (possible solutions include: better incentive structure. Performance based reward system, better margins to sales people, etc.)

Candidate should cover this next:

Data collection in real life

Possible questions to guide the interviewee:

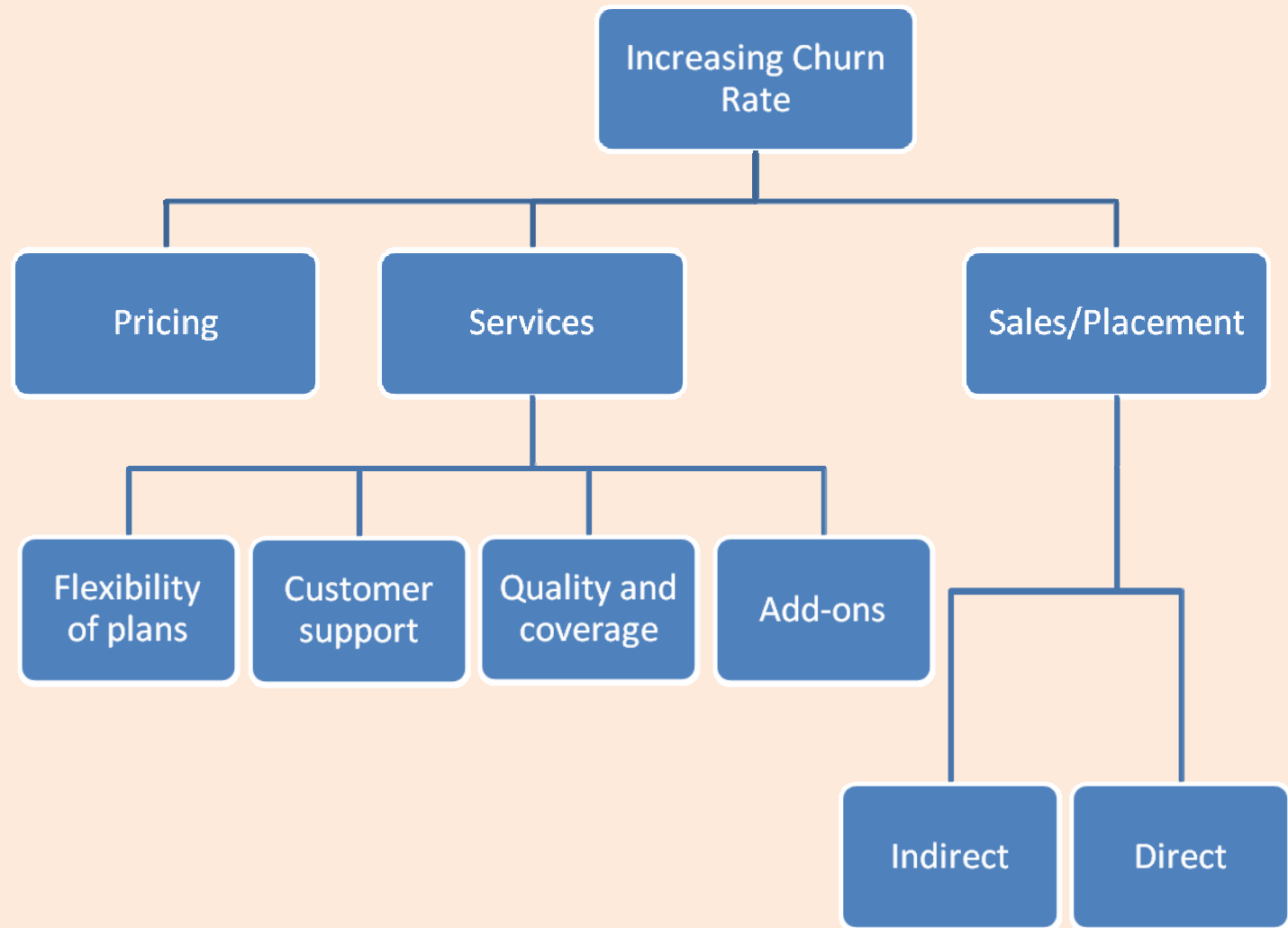
- What are possible sources of data? (Customer surveys, research reports)
- Are customer surveys good enough? (they can be biased)
- How would you analyze the data and account for the bias? (use regression tools)

A solid interview will address these following areas

A solid interviewee will try to bring his own experiences into the case. Like: If he/she were to but a cellular phone then what factors will affect the decision making process.

Case Details (Page IV of IV)

Possible structure
for the case:



Problem statement (Page I of VI)

Company: Booz
Year: 2008
Round: I
Case Name: Regional Health Plan

Problem Statement - Narrative

Your client is a regional health plan. Within the last 2 years their profits have gone down by 15%.

Their biggest business division provides insurance solutions to the customers. As part of the business model, they collect premiums from the customers and pay part of the medical costs.

Objectives: The CEO of the health plan has hired you to find out the reasons behind declining profits and recommend solutions to stem the problem.

Guidance for the interviewer

- Not much numbers involved in the case.
- The case can be approached by using the profitability framework.

Case Details (Page II of VI)

Candidate should cover this first:

Revenue side

Provide the information when requested:

Company:

- Smaller player in the market
- Charges price premium as compared to closest competitors
- Differentiates on customer service, easy-to-navigate system, more insurance options
- Operates in the north-eastern US only, does not have any plans for expanding into other markets
- The number of customers are growing slowly and steadily but their demographics are changing

Market & Competition:

- The overall market for insurance is growing at 5% – 6%
- The competition is average with many players in the market.
- The competitors are other regional players as well as national players also.
- There is no one dominant player in the market.
- The competition's products and services are cheaper than our client's products

Focus on how the interviewee structures the problem using the information provided above. The interviewee should be able to correctly guess that:

- There is no scope of playing on prices
- Since the number of customers is also increasing, the problem is not on the Revenue side

Case Details (Page III of VI)

Candidate should cover this next:

Cost side

Provide the information when requested:

Costs:

- Ask the interviewee to draw a value chain for the costs in the insurance industry.
- The interviewee should cover the following buckets: Customer Acquisition costs → Customer Support costs → Pay-Out costs → Back office costs
- Customer acquisition costs are normal and comparable to similar sized competitors
- Customer support costs are more, given the company's focus on customer service. But these costs are expected to be higher and it is difficult to cut corners.
- Pay-out costs are mostly standard in the insurance industry
- Ask the interviewee to further break down the back office costs. A good list would be: IT costs + administrative costs + supplies and equipment costs
- Ask the interviewee to further break down IT costs. A good break up would be: Employee compensation + Software + Hardware
- Ask the interviewee about the drivers for the Employee compensation bucket. Two major drivers would be: Salary + Number of employees.

Case Details (Page IV of VI)

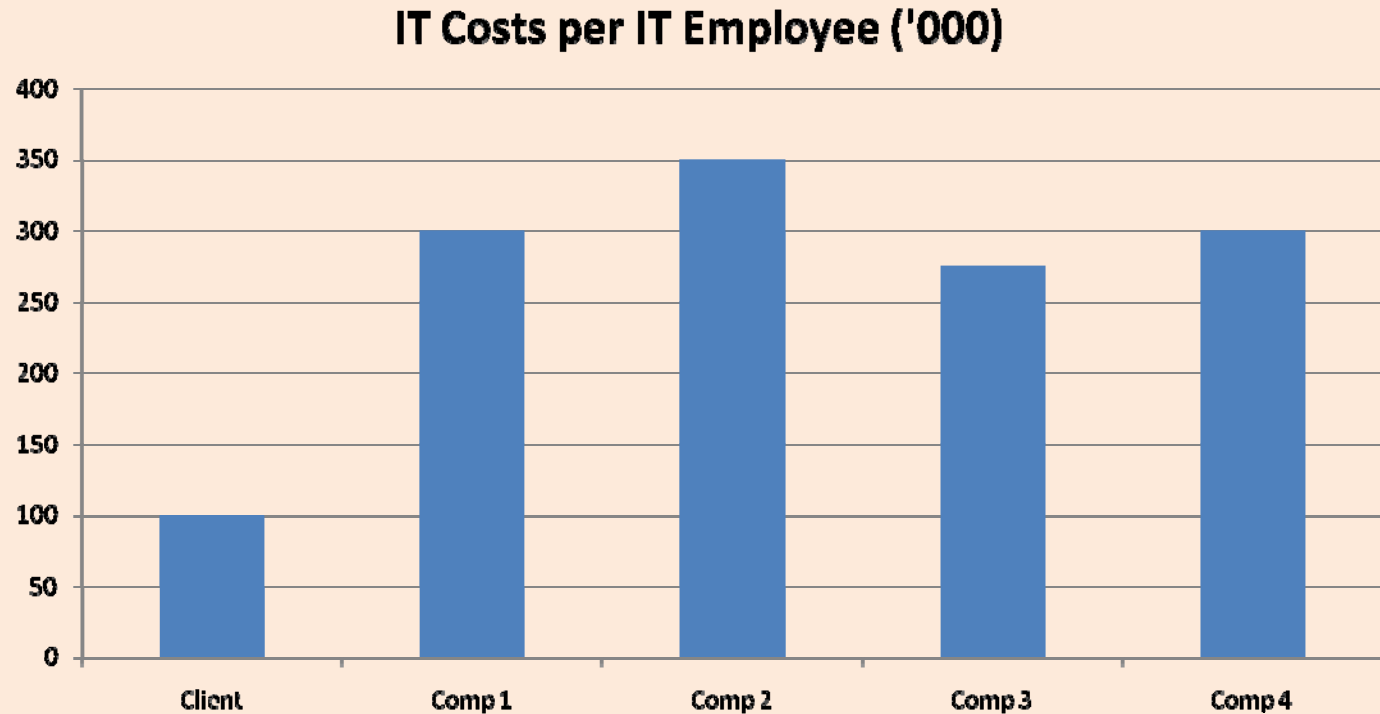
Candidate should cover this next:

Cost side

Show the following graph to the interviewee and then ask the following questions:

- What can you make out from the graph?
 - Answer: Client's IT cost per employee is very low as compared to the competitors
- Is that good or bad?
 - Answer: Depends. If the number is low because the client has low IT costs then it is great. However, if the number is low because the client has more employees then it might be bad.

IT Costs per IT Employee



Case Details (Page V of VI)

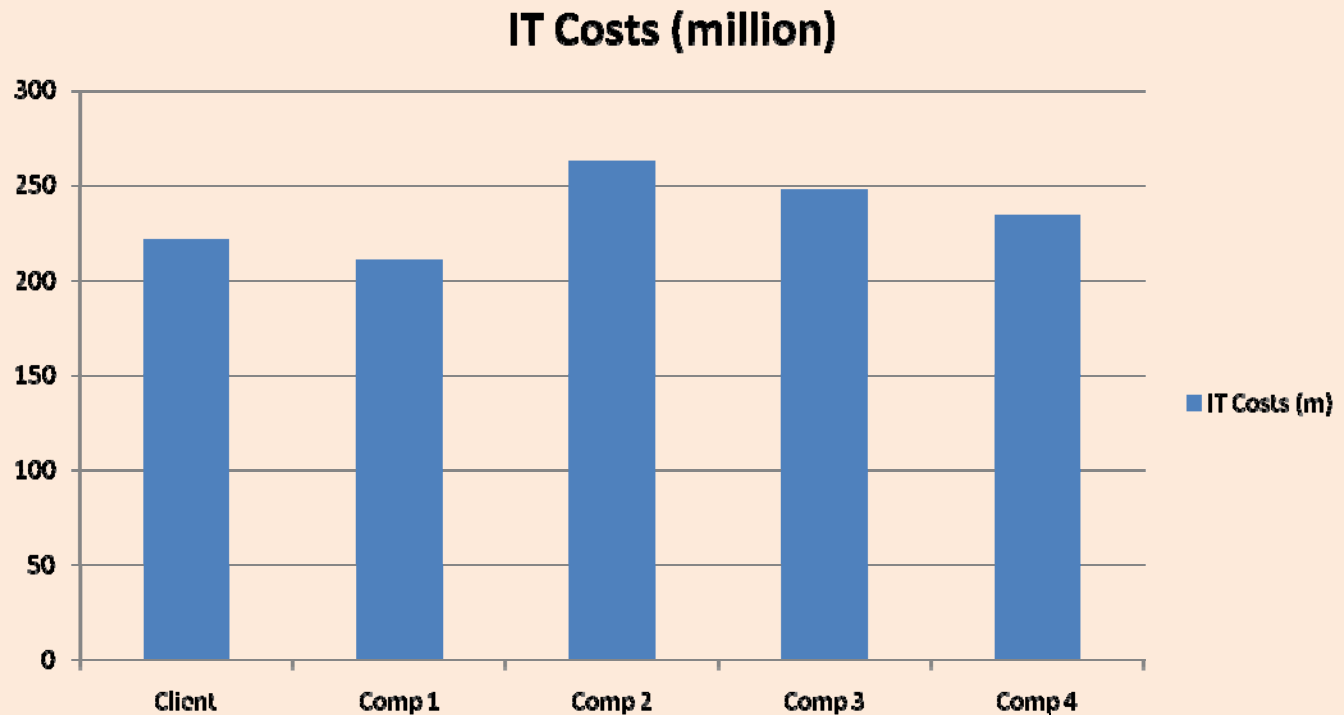
Candidate should cover this next:

Cost side

Show the following graph to the interviewee and ask the following questions:

- What do you make out from the graph?
 - Answer: Client's total IT costs are comparable to the closest competitors
- What does that tell you?
 - Answer: This means that the number of IT employees at our client is very high as compared to the closest competitors.

Total IT Costs (m)



Case Details (Page VI of VI)

Candidate should cover this next:

Reasons and Recommendations

Ask the following questions to the interviewee:

- What do you think can be the reason for such a high number of employees?
 - Possible answers include:
 - lower efficiency of the employees,
 - larger number of current projects,
 - lack of cross-functional synergies (different teams doing similar

work)

- What can be done to mitigate some of the potential problems that you have listed?

Problem	Correction
lower efficiency	change incentive structure, better project management
larger number of current projects	review projects and put non-critical ones on hold to reduce short term costs
lack of cross-functional synergies	Institute/Change “new project” review procedure

Problem statement (Page I of III)

Company: Booz
Year: 2008
Round: I
Case Name: NGO Effectiveness

Problem Statement - Narrative

Your client is a NGO (like Red Cross) having operations in around 100 countries. The two major services they offer are Disaster Relief and Child Sponsorship.

Objectives: The NGO leadership feels that the 'end results' are not meeting the set objectives.

Specifically,

1) They are not able to raise enough funds to meet their targets.

2) They are not able to utilize the funds they have effectively.

They have asked for your help to find ways to raise more funds and to identify the areas where they can cut some costs.

Guidance for the interviewer

- This is a slightly open ended qualitative case.
- The interviewee can treat the case as a Revenue/Cost problem. Finding ways to increase revenue and finding ways to decrease costs.
- The disaster relief services are provided locally in the area of a disaster (like Asian Tsunami, earthquakes, etc.)
- The child support services are where people donate money to support a child in a poor country, in terms of food, lodging, schooling, skills workshops, etc.

Case Details (Page II of III)

Candidate should cover this first:

Ask the interviewee to come up with potential revenue streams and cost centers for a NGO. Let them brain storm on the possible streams and see if they can weave they own experiences into the case.

Provide the following information when required:

Revenue streams: The donors are classified as Individual donors (small players) and Rich Individual + Institutional donors (big players). Government help can be ignored. The revenue streams can be also be broken down by Disaster Relief operations and Child support services.

Cost streams: There are two major cost centers for the NGO: Administration and Training. Administration can be further divided into sub centers as Back Office, Transportation, Recruitment and Advertising.

Candidate should cover this next:

Explore Revenue Streams

Information to be provided when asked:

- General donations to the NGO go to a pool from where they are divided between Disaster Relief and Child Support, as required.
- People also make specific donations for Child Support services.
- Who draws in more donations? Currently, two-third of the revenue comes in for Child Support activities. Hence, direct the interviewee towards that.
- Who are the major contributors? Most of the money comes from individual donors.
- Individual donors are more sensitive to the results on their donations.
- Ask the interviewee for different approaches to reach out to individual and institutional donors. There are no right or wrong answers.
- Example of possible answers: People tend to donate more when they can see that their donations are producing results. Hence, develop a system where the donors can be in touch with the child they are supporting. Bring in some ownership and form emotional bond. Use that to advertise and attract more individual donors.

Case Details (Page III of III)

Candidate should cover this next:

Explore Cost Centers

Information to be provided when asked:

- The interviewee should be able to guess that the Administration activities are the bigger cost center and hence focus on that branch.
- Within Admin, Back-office operations form the largest chunk.
- What are some of the major Back office activities for an NGO that you can think of? Answer should include managing communication and managing money being distributed.
- What are the things you would look at for understanding these activities? Answer should include structured thinking on the lines of People, Processes, and Technology.
- Each country's operations have their own spreadsheet-like software for accounting.
- No common framework within the organization to manage communication.
- How can you leverage technology to improve accounting and communication. There are no right or wrong answers.
- What are the benefits that can be achieved by leveraging technology? Possible answers: Common framework within the organization, reduced duplication of efforts, lesser training costs, more transparency, etc.

A solid interview will address these following areas

A solid interviewee will try to bring his own experiences into the case. Like: Where do NGOs get their money from? What are the most common cost centers for NGOs which have world wide operations?

Problem statement (Page I of IV)

Company: Bain
Year: 2008
Round: II
Case Name: Tire Manufacturer

Problem Statement - Narrative

Your client is one of the largest tire manufacturers in the USA. The current market for tires is saturated and with current pressure on car prices they are looking for new markets.

- There is a new market for rubber roofing. There is limited capacity in the current setup
- There is a competitor with a much larger capacity and economies of scale

Objectives: What should our client do?

Guidance for the interviewer Information to be provided upon request

1. This is a good case on how to weigh pros and cons.
2. The critical point is to find out who the customers are in the construction industry.
3. The next point is to find out how this new rubber roofing is better than the existing roofing, the rubber roofing is cheaper to install and lasts longer. Its thickness is made to have the same effect as the existing gravel roofing.
4. The client has revenues of over 3 billion. This market is only 5 million per year.
5. The competitor who can make the roofing cheaper and has more capacity.

Case Details (Page II of IV)

Candidate should cover this first	1. Customers (who are they, what drives them) 2. Product (how is it better than the alternative) 3. Competitors (what advantages/disadvantages) 4. Company (capacity and investment)
Candidate should cover this next (math section)	Option 1: The math in this case is simple, it compares the total cost of ownership (use that phrase a lot) of the rubber alternative to the gravel one. Option 2: The math can also be used to determine the market size (which you should do!) and then conclude that the company can only serve a small portion of this market with the existing capacity. The market size, you should know that many houses will not switch, apartments have only one roof for many houses etc. Just say these things
A solid interview will address these following areas	1. The key is that the 3C will not help unless you manage to incorporate the product in great depth 2. Remember that this is a SECOND round case. Just getting the info is not good enough, you need to build on the info and say what it MEANS to the client
Value Chain if any	1. No particular value chain but when comparing the benefits of the rubber to the gravel alternatives you should look at a process

Solution summary (Page III of IV)

Questions for the interviewee

Question #1 – *What should our client do?*

Question #2 – *should we invest in more capacity?*

Question #3 – *Where do you see this industry in the future?*

Note that this question may come somewhere in the middle of the interview not the end!

Sample Solution

Poor Response #1 – *Wait and watch*

Good Response #1 – *Just make the small amount*

Excellent Response #1 – *try and capture the market*

Poor Response #2 – *no this is not our core business*

Good Response #2 – *no we cannot compete*

Excellent Response #2 – *maybe if we can leverage our expertise in rubber*

Poor Response #3 – *Cant say, need more data*

Good Response #3 – *Will grow*

Excellent Response #3 – *not only will new houses be rubber roofed, the existing homes may switch over so it has tremendous potential*

Solution summary (Page IV of IV)

Math Questions for the interviewee

- The current roofing is gravel.
- Costs \$20 per ton and has a layer of 6 inches.
- The time it lasts is 4 years.
- The rubber roofing costs \$30 per ton and has a layer of 4 inches and lasts for 10 years.
- Bonus math. Only if the candidate asks.
- The time taken to apply the Gravel is 30 min per square foot, while the time taken to apply the rubber is only 10 min.
- This is if the candidate can guess time to apply will be a factor.

Problem statement (Page I of II)

Company: Bain
Year: 2008
Round: II
Case Name: Dominos Pizza

Problem Statement - Narrative

Your client is one of the largest pizza companies (Dominos). They have seen the population of a town change over the years and want to know how best to market their products to maximize profits

Objectives: What should your client do?

They want to identify where they can open new stores.

Guidance for the interviewer Information to be provided upon request

1. This is a good case on how look at market entry and strategy.
2. The critical point is to find out what the business model is and what the pricing strategy is.
3. The next point is to find out how the relationship between the client and the franchisers works.
4. There is some math on how many shops to open.
5. There is some strategy on what customers consume pizza.
6. The tricky math is elasticity of demand and revenues. With lower prices, more revenue to franchise but cannibalization.

Case Details (Page II of IV)

Candidate should cover this first	1. Customers (who are they, what drives them). 2. Product (how is it better than the alternative), home delivery 3. Competitors (what advantages/disadvantages), 4. Company (this is the key of the case, you need to get that it is a franchise model)
Candidate should cover this next (math section)	Option 1: The math in this case is tricky but that is not the key. They will ask you if the price is elastic or not. This is a red herring. They will then give you elasticity of demand (say 20% discount means X% increase in sales) Then you will calculate that the pricing should be lower for a single store to make more profits. But this is the problem. Lesser pricing means less profit for Dominos and there is cannibalization between stores. Even if a franchise can go lower price they shouldn't.
A solid interview will address these following areas	1. The key is understanding that what is good for the franchise (owner operator) is not necessarily good for the client (dominos) 2. Remember that this is a SECOND round case. Just getting the info is not good enough, you need to build on the info and say what it MEANS to the client
Value Chain if any	1. Now this an area where the value chain becomes critical. 2. Identifying the franchise model and then the turnkey approach

Solution summary (Page III of IV)

Questions for the interviewee

Question #1 – Where should they open stores?

Question #2 – should franchises allowed to increase price?

Question #3 – Where do you see this industry in the future?

Sample Solution

Poor Response #1 – Wherever they want as it's a franchise models.

Good Response #1 –they should analyze the customer segments, college students medium income etc.

Excellent Response #1 – they should franchise 80% and keep 20% of the stores in the high rent areas. The franchisees cannot afford rent in downtown but our client will have an image.

Poor Response #2 – yes. They make profit

Good Response #2 – no we should have standardization

Excellent Response #2 – not in the short run but we should investigate why they want to lower price and if it makes sense, share the best practices

Poor Response #3 – Cant say, need more data

Good Response #3 – Health food is growing

Excellent Response #3 – the entire industry for pizza can change to lower rent. Move from sit-in restaurants to more home delivery and packed food in the office style. We should understand how we can incentivize franchise owners.

Solution summary (Page IV of IV)

Math Questions for the interviewee

1. 20% drop in price increases volume by 30%
2. Price elasticity = 1.5 (or – 1.5 to be precise)
3. Assume all fixed cost in the franchise remain unchanged. Assume all variable costs are in proportion to quantity
4. If revenues were $\$10 * 100$ units, Current revenue = $\$1,000$
5. The new revenues = $\$8 * 130$ units = $\$1,040$

This is a 4% increase in revenue.

Fixed costs remain but are divided across more units.

Math Questions for the interviewee

So more profits. Margin (based on variable costs) remains unchanged.

Risks: There may be some cannibalization. And maybe other franchises cannot expand (capacity) without adding fixed costs.

Problem statement (Page I of II)

Company: ATK
Year: 2007
Round: I
Case Name: Baked Snack Goods Manufacturer

Problem Statement - Narrative

Your client is a baked snack goods manufacturer (i.e. cookies, crackers, etc) based in the U.S. It operates using a direct store delivery model. They operate and own their own trucks driven by company employees. When delivering, they deliver directly to the supermarket floor where employees park at the loading dock, unload goods, and place it on the shelves. For the most part, employees at supermarkets do not touch the goods at all. This model is industry standard.

Objectives: Your client has done a cost benchmarking analysis and has discovered that their distribution costs are higher than competitors. What is going on?

Guidance for the interviewer Information to be provided upon request

- Two Key Cost Areas
 - Warehousing Costs
 - Fixed Costs: Land, Maintenance, O/H
 - Variable Costs: Inventory Holding Costs, Labor
 - Trucking Costs
 - Fixed Costs: Trucks, Maintenance, O/H
 - Variable Costs: Fuel, Labor
- Company has 15 trucks
- Truck driver utilization is approximately 70%, which is lower than the average in the industry
- Grocery Stores have a four hour delivery window requirement for the client
- The delivery window requirements are normally distributed with most of the customers requiring that deliveries be made in the hours between 8:00AM and 1:00PM
- Client's direct labor wages paid are the same as the competition
- Truck drivers have 10 hour shifts
- Direct Labor Costs on the trucking side are higher than average

Case Details (Page II of II)

Candidate should cover this first	1. Identify major cost buckets: Warehousing and Trucking
Candidate should cover this next (math section)	Option 1: Why are the direct labor costs on the trucking side higher than average? Option 2: Why is utilization lower than the industry average?
A solid interview will address these following areas	1. Labor costs are higher than average for two potential reasons : 1) The wages being paid by this client are higher than the market wage and / or 2) The client has more employees than is necessary (i.e. they are not utilizing the available labor hours properly) 2. Utilization is lower than average Three reasons: 1) Demand is not predictable, 2) The process for loading the trucks is not efficient (i.e. truck drivers are waiting for the trucks to get loaded), 3) Supermarkets have a 4 hour delivery window. Outside of this window, the truck driver's time is not used efficiently, and thus, the drivers AND trucks are idle.

Solution summary (Page I of I)

Questions for the interviewee

Question #1: What should our client do?

Sample Solution

Excellent Response #1: There are internal and external methods for addressing this issue. Internally, the client can begin hiring part time drivers to manage the delivery of goods to supermarkets. These part time drivers can be hired to specifically work during the delivery windows. This can help with the client's direct labor costs. The client though, should be cognizant of the risks of doing this though. Hiring part time drivers can potentially compromise quality of delivery and potentially drive up employee management costs. Externally the client can attempt to negotiate with the customers to try to flatten out delivery window requirements such that the client's labor/equipment resources can be used more efficiently. Another alternative to easing the pain is to attempt to sell the idle truck driver time to other customers. An example of this would be to business that would like to distribute newspapers since delivery time for this type of business would not fall within the 8AM – 1PM segment.

Problem statement (Page I of VI)

Company: McKinsey
Year: 2008
Round: II
Case Name: Retailer Growth Strategy

Problem Statement - Narrative

Your client is a big retail chain in United States, which owns the entire distribution and retail supply chain. The client has seen sustained profits for last several years and is now looking for several growth alternatives in terms of cost reduction and increasing market share. Your consulting firm has been brought in to prioritize the various alternatives and decide the order of execution.

Background (Provide only if asked)

The retail stores sell drugs for normal usage such as cough and cold medicines etc. (non-prescription based). Industry has seen some shift towards being green and client has not yet done that.

Objectives:

- Present a rough timeline for execution
- Identify risk and impact of executing these strategies

Guidance for the interviewer

Information to be provided upon request

1. As with most 2nd round cases, this is a totally qualitative case.
2. The interviewer should present the strategies to the interviewee and should probe the interviewee for his thought process.
3. There is no right way to solve this case, but organizing the various factors to analyze in a tabular format and then plotting the various growth strategies in a 2 * 2 matrix is the best way.
4. If the interviewee does not plan to use a 2 * 2 matrix, encourage him/her to use one after the preliminary analysis.

Case Details (Page II of VI) – Sample Solution Format I

Provide these strategies to candidate	1. Implement lean manufacturing processes 2. Reorganizing Sales force(Incentives for sales force) 3. Change packaging of products in retail stores(recyclable material) 4. Launch new products in market 5. Change Sales Floor layout
Candidate should cover this next	The candidate should come up with a tree structure to analyze the various growth strategy options and then create a table with all the 5 growth strategies and the factors that should be considered for evaluation. Once candidate has covered most basic attributes for evaluation, ask him / her to rate the various attributes in a rank scale order. A sample tree and an evaluation table is provided in the next page for comparison.
Next Steps : 2 * 2 Matrix	Once the candidate has sufficiently addressed the various evaluation criteria, he should then next plot the various growth strategies in a 2 * 2 matrix with the 2 dimensions of risk and impact. A sample 2 * 2 matrix is provided in next section.
Focus Areas	The key focus area in the 2 * 2 matrix is the fine balance between risk and impact. This is where the candidate should ask questions about the current state of the market, company and competition. The interviewer should mention that some green initiatives in industry has been causing some trouble with some packaging for the drugs.

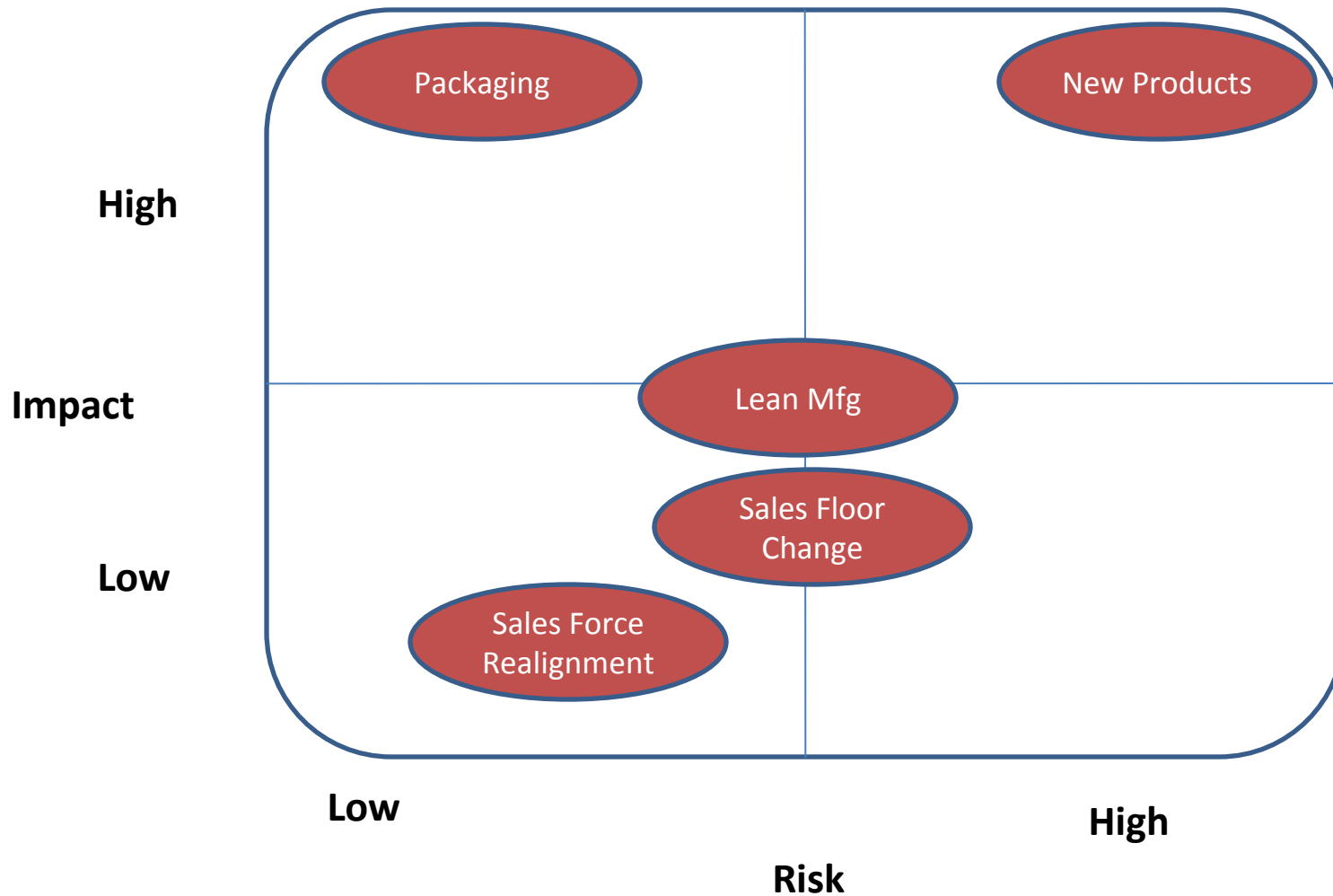
Sample Tree (Page III of VI)



Evaluation Table (Page IV of VI)

Strategies	Risk	Top Line	Bottom Line	Long Term Vision / Brand Strategy	Competitive Edge
Lean Mfg	Medium	None	High	Low	Medium
Sales Force Realignment	Medium	Low	Low	Low	Low
Packaging	Low	Low	Medium	High(Go Green)	High
New Products	High	High	Medium	Medium	Medium
Sales Floor Change	Low	Medium	Low	Low	Low

2 * 2 Matrix (Page V of VI)



Solution summary (Page VI of VI)

Questions for the interviewee

Question #1

What are some of the risks with execution order Strategy?

Question #2

Is this a long term sustainable solution?

Question #3

How can the retail store leverage its dominance in the supply chain?

Question #4

What if the competition also copies green initiative?

Sample Solution

The main purpose of this case is to understand the thought process of a candidate when an ambiguous business problem is encountered. The overarching issue with the company right now is the focus on green initiative by the industry, and that is why the impact of the packaging change to recyclable material is a medium impact strategy. A sample order

of execution is as follows:

1. **Packaging Change:** Start “Go Green” initiative
2. **Sales Floor change:** Reorganize the drug products to emphasize new packaging
3. **Sales Force realignment:** Focus on the new green brand
4. **Lean Manufacturing:** Long term sustainable cost advantage
5. **New Products:** Product differentiation

AGRICULTURAL EQUIPMENT MANUFACTURING

Your client is a large agricultural equipment manufacturer. Their primary product line, farming tractors, is losing money. What questions would you ask of your client to help them solve their profitability problem?

Answer: Agricultural Equipment Manufacturer

A: It is unlikely that there are too many players in this market. You might want to start off by asking how many competitors there are. Suppose the answer is that there are two direct competitors.

What is your client's market share relative to their competitors (your client has 40% of the market, competitor #1: 30%, competitor #2: 15%, with the remaining 15% belonging to many small manufacturers.)

What are the market share trends in the industry? (Five years ago, your client had 60% of the market, competitor #1, 15%, and competitor #2, 10%. Obviously, your client has lost significant market share to its two competitors over the last few years.)

Do all three competitors sell to the same customers? (Yes)

How is your product priced relative to your competitors? (Your client's product is priced higher than the others.)

Has this always been the case? (Yes)

Are the products the same? (Essentially yes, they all have the same basic features. Of course, tractors are not commodity items and a few differences do exist.)

What are the differences that allow you to charge a premium for your product? (Your client has a strong reputation/image of quality in the market and the market has always been willing to pay a premium for that reputation because it meant they would last longer and need less maintenance. This can be critical for some farmers because they cannot afford to have a piece of equipment break down at a critical time.)

Are sales revenues down? Are sales quantities down? (Yes)

Is the price down? All costs the same? (No, in fact both the price and costs are up.)

Have fixed costs increased? (No, material costs, (variable costs,) have gone up out of sight, and the client has no answer as to why material prices have gone up so staggeringly.)

Do you manufacture your tractor or just assemble it? (Primarily an assembly operation.) Finished part prices have gone up? (Yes)

Raw material prices for your suppliers? (I don't believe so)

Have labor costs increased for your supplier? (No)

Have you changed suppliers? (No)

Why are your suppliers charging you higher prices for the same products? (Well, they're not, the prices have increased as a result of our product improvement efforts. We've tightened tolerances and improved the durability of our component parts.)

Why do you make these improvements? (Because we strive to continue to sell the best tractors in the world.)

Are your customers willing to pay for these product improvements? (What do you mean.)

Are your customers willing to pay a marginal price which will cover your cost of implementing these improvements? (I don't know, I guess we assume that they will...)

It turns out that prices have been raised to cover the costs of these improvements, but customers do not value these improvements unless they are essentially free --so sales are down. The client needs to incorporate a cost/benefit analysis procedure into its product improvement process. Don't forget though, that you must consider the long-term effects of these decisions.

TERMITE PESTICIDE

You have been hired by a pesticide application company (think the Orkin Man) to evaluate the feasibility of adopting a new form of termite pesticide. Your job is to recommend which product the company should use and how they should market their choice.

Current Product

The current product is a two phase operation a technician places baiting boxes into the ground around the client's house. After two weeks the technician returns to see if the termites have eating the wood bait. If there are signs of termites, the technician will fill the baiting boxes with "laced" wood which will effectively kill the colony.

New Product

The new product is a liquid application that is applied (sprayed) onto the foundation of the house regardless of termite infestation.

Notes on the two products:

- Both products are equally effective
- Both products are equally safe

Price/Cost – Baiting is more profitable for a first year application.

Treatment

Type	1st Year		Renewal	
	Cost	Price	Cost	Price
Liquid	\$750	\$1,000	\$100	\$200
Baiting	\$1,100	\$1,500	\$250	\$300

Notes on Profit

- Here the interviewee should calculate the profit and realize baiting is more profitable in the first year. (\$400 opposed to \$250)
- What you need to do is make them think about the renewal aspect... i.e. the customer
- There are no fixed costs associated with the liquid treatment

Customers – Renewals and Profits – Overall Liquid is more profitable

Customers renewal rates diminish from their initial application, the interviewee should calculate the contribution margin of renewal rates by multiplying the percentage by the profit of a renewal.

Customer Renewal Rates

1st Year	90%
2nd Year	80%
3rd Year	70%
4th Year	60%
5th Year	50%
6th Year	0%

Total Profit with Renewal

Liquid	\$600
Baiting	\$575

Customers – Product Preferences

In a customer survey we found the order of their preferences.

1. **Efficacy**
2. **Safety**
3. **Price**

Notes on Customer Preferences

1. Since the efficacy (effectiveness) is the same for both this is not a concern
2. There is a perceived safety associated with baiting opposed to spraying (liquid), so the client is going to have to educate its customers that both applications are equally safe
3. The liquid application is less expensive for the customer for the initial application and for renewal

Competition:

There are no local competing companies at the moment, but companies in adjacent towns are offering the liquid service at the same prices you are considering.

Notes:

The interviewee should understand that the competition will offer the product if they do not.

Overall Recommendation:

- Initially the company should offer both products to meet customers who prefer safety over price and price over safety.
- The client should spend money on educating consumer that liquid application is as safe as the old baiting method
- The client needs to monitor competition to ensure the dominate position in town

TELECOMMUNICATIONS DIVERSIFICATION

A Baby Bell company is interested in diversifying into other areas besides telecommunications. They are considering entering the market for electronic home security systems. Would you recommend that they do so?

Suggested frameworks:

Use an industry attractiveness framework, such as Porter's Five Forces, to determine whether this is a business you want to be in, or at least to determine what kind of returns you can expect to achieve. then, use the value chain to look at where value is added in the home security business. finally, once you feel you understand the market, determine if the core competencies of the Baby Bell are likely to match the demands of the home security markets.

Interviewer Notes:

The company is a holding company. They have previously made unsuccessful forays into software and into real estate.

The home security business is highly fragmented. The top five players in the industry generate less than 4% of the total industry revenues. This implies that the industry largely consists of small, regional companies.

10% of all residences currently own an electronic security system.

This is some sense a razor and razor blade sort of business. The economics are:

<u>Item</u>	<u>Retail Price</u>	<u>Cost / Margin</u>
Equipment and Installation	\$500 - \$1,500	0-10% margin
Monthly Service	\$20 / month	\$5 / month

What strengths / competencies of the Baby Bell Company are useful in this market? Consider: Installation expertise, operator services, transmission system (phone lines)

It turns out that the "expensive home" segment of this market is saturated. Growth has been slow in recent years.

Price sensitivity is unknown in "moderate-priced home" segment.

The conclusion is that this business is a reasonably good fit for the company, but that more market research needs to be done to assess the growth and profit potential of each segment of the market.

TECHNOLOGY IN BANK

A large bank has been around for 120 years. 30 years ago, they invested heavily in technology and automation of majority of the transactions. However, with the changing business conditions and the advent of the new technologies like wireless and Internet, they realize that they need an overall of their technology.

The CIO has commissioned you as a consultant to estimate the cost of porting their current infrastructure into a web based state of the art system. The ballpark cost is to be estimated in 30 minutes. How would you approach this? In addition to cost, also come up with some key guiding principles of the estimation.

Information to be given when asked

- The client is not interested in reengineering the business processes, at this time. Focus only on the software cost and the technology overall cost.
- The legacy applications have 1 million lines of code.
- Cost for porting the application can be taken as \$200 per day per developer.

Suggested approach

- As with most technology overhaul case, the candidate should be able to identify that there is a possibility of business process improvement as well, in addition to the technology changes.
- Identify the various phases of this project. Software development is just one of the phases. Other phases could be Requirements gathering, business analysis, testing, and implementation, among others.
- Could start by calculating the cost of only software development. With 1 million lines of code and say 25 lines per developer, you need $1,000,000/25 = 40,000$ man days. With \$200 per man-day, the cost comes to $40,000 \times 200 = \$8$ million.
- Make reasonable assumptions about cost of other phases. One approach is to assume that you will spend approximately the same amount (\$8 million) in business analysis, managerial overheads, requirements gathering etc, and another \$8 million in say testing, implementation phases, bringing the total to around \$24 million.
- Must also mention some guiding principles and other factors to consider for this project. Some of the factors are:
 - Outsource vs. in-house development
 - Phased implementation vs. Big Bang
 - Using industry tools, practices and reusing available frameworks
 - Cross functional teams for defining business requirement

The key to this case is the ability to identify phases in a software development project, and the issues involved. Reasonable assumption can be made about most numbers involved.

SUPER REGIONAL BANK

You have a have recently been assigned to a project with one of the nation's super regional banks. The bank is one of the top 10 largest retail banks in the country. Like most banks in its class it has branches in 8 geographically contiguous states.

Your client has recently concluded that the old "local branch" way of business is no longer viable. Typically, this bank has canvassed its territory with small free-standing branches; however, the new age of electronic banking and commerce is changing all of that.

They are considering replacing many branches with Calling Centers. Calling Centers offer both live and phone automated services that may be accessed by phone. The new Centers would offer virtually all of the services currently offered through local branches plus some additional things.

The question to you is: how would you go about setting up the engagement to determine the viability of this new concept? Specifically, what kinds of things would you investigate? And what hypothesis would you form?

Possible Solution:

This is a very open broad-brushed case. There certainly is no right answer; however this type of case occurs frequently. The following is a guideline of some things you should probably consider:

Market analysis: What kinds of customers would be attracted to this no service? What kinds of customers would be turned off? (Hypothesis: younger people would be heavier users and more attracted than older) Of the people attracted to this new service, how profitable are they? How profitable are the people who are turned off by this service? (Hypothesis: older people have more money and thus are more profitable)

Revenue: What types of new services could be added to increase revenues? Automatic bill payment, Fund transfer, etc.

Cost Savings: How much would it cost to establish a Calling Center and what are the risks involved? Do we have the expertise in-house to do this? How many branches could we close? Can we cut down on traffic to existing branches - thus requiring less tellers?

Summary: It probably is best setup as a cost benefit analysis. The number of new customers times the expected revenue from them plus the additional revenue generated by potential new services plus the cost savings must outweigh the forgone revenue generated by the customers you end up driving away.

SNACK FOOD COMPANY

A large salted snack food company has steadily been losing market share over that past two years, from a high of 20% to the current level of 18%. Profits as a percent of sales, however, have been growing. What could be causing this?

Additional Information to be divulged gradually:

The size of the total salted snack food market has grown from \$15 billion to \$17 billion during these two years; the interviewee's conclusion should be that the client's total dollar sales have actually grown, but not kept pace with the market. The product line of the client has not changed over this period.

The costs for the client have changed over this period: (% of selling price)

	Current	Two years ago
Raw Ingredients:	28%	26%
Conversion costs:	24%	24%
Distribution:	8%	9%
Marketing:	16%	18%
Sales force:	7%	9%
Pre-tax profit:	17%	14%

The total sales force was cut to reduce costs, although the same number of outlets are still covered by this sales force. The changes in the marketing budget come from reduced trade promotions.

The products are mostly sold through large grocery store chains and convenience stores. The sales force generally visits each customer at least once per quarter. Promotions usually occur at the end of each quarter. Grocery stores and convenience stores require some type of promotion to grant valuable end of aisle displays or advertising space.

The largest competitors are two multinational consumer products companies that feature complete lines of snack foods. Their sales forces are regarded as the best in the industry. Together, these two companies have 55% of the market.

Solution:

The data show that the greatest change is in the sales force numbers. It turns out that the company went on a cost-cutting spree over the past two years. The sales force was drastically cut and the commission scheme was reworked. The marketing expenditure was also decreased. Most of the reduction came from trade promotions. The product is sold through the same channels as previously: large grocery chains and convenience stores. These channels are traditionally driven by periodic trade promotions. The reduction in trade promotions brought about a loss of shelf space, which has directly led to the decrease in market share. Also, the product line has not changed in the past two years in a product category where new products and line extensions are routine. In addition, the market has been growing, indicating a missed opportunity for new products in the market. Lastly, the increase in profitability has resulted from the lower costs, but may not be sustainable.

SHIPPING AND TRANSPORTATION COMPANY

Our client, \$3 billion transportation and shipping company, is facing declining profit margins since the last five years. The client is only experiencing 3%-5% profit margins but wants to return to 10% margins. Historically the client is a first mover in the industry and has had rapid growth.

Information to be given when asked:

- The client is an international company with large distribution centers
- Kinds of shipments handled are overnight, regular for both domestic and international destinations
- Market is mature with a growth rate of 4-5% per year.
- Revenue growth of the client is consistent with the industry
- Competition: Three major competitors, have similar operations but beating us on price.
- Revenue is not an issue here. Focus on the cost.
- High administrative costs in several support departments like HR, IT, engineering.
- Mature market, fixed costs kind of constant, no new investment in warehouses, trucks, planes etc.
- Ask the candidates about what could be the SG&A costs, pick up a department like IT or engineering and ask how costs could be reduced in these departments.
- If asked by the candidate, there are 1500 employees in the IT department. Some 300 engineers work on scheduling and shipping algorithms.
- If the candidate suggests layoffs as a possible cost reduction strategy, ask him on how he will communicate it to the CEO. The candidate should be able to relate any such suggestion to the overall objective of the CEO (10% or more growth in profitability) and convince the CEO that such measure is related to the growth objective

Suggested approach

Though not explicitly stated, the company is like FedEx or UPS and this should help the candidate visualize the case situation.

- The candidate should be able to focus on the profitability equation and competition.
- Revenue growth is in line with that of the industry so costs are potential problem.
- Should be able to break down the costs into fixed and variable and realize that since this is a mature industry and the company is not exactly investing in fixed assets at this time. Variable costs are too high, specially the SG&A and support department.
- Some cost cutting measures should be discussed. For example in the IT, they could consolidate the hardware, in-house development of only the complex projects while outsourcing the more standard kinds of development functions. Cost cutting may entail some layoffs, realignment of staff duties.
- It is important to relate any cost cutting measure with the growth objectives.

SHIPPING COMPANY - CRM

Your client is one of the biggest overnight shipping companies. (e.g. FedEx) They are considering building an in-house CRM solution. They have 3,000 people in IT department and expect \$15 million spending for the implementation project. Maintenance cost would be 50% of implementation cost. You are dispatched to this company as a consultant to suggest the possibility of purchasing CRM package. How would you structure your analysis?

Candidate: I would look through at the three major areas – cost / benefit / external issues.

Interviewer: OK. Go ahead.

Candidate: First, cost issues. What would be the anticipated implementation cost of purchasing CRM package?

Interviewer: It would cost \$5 million to purchase the package. Maintenance cost would be 15% of the price.

Candidate: What about the related consulting expense?

Interviewer: Right. Consulting fee would be \$7.5 million.

Candidate: Then total cost of purchasing would be \$12.5 million with 15% maintenance cost.

Interviewer: Right. But there might be a hidden cost occurring when you buy the package. What would they be?

Candidate: First of all, there would be education expense – employees need to learn the package. At this time, I couldn't think of others....

Interviewer: Well, there would be compatibility issues – additional cost will be incurred if the package is not connected well with existing legacy system.

Candidate: That's right. I missed that but it's an important point. So it would be hard to say that buying the package has a cost advantage. Then I'll look at the benefit side. I would assume that In-house CRM system would provide better fit to existing business process of the client. Is it okay to assume that?

Interviewer: Well, it is generally true.

Candidate: Is it okay to assume that the IT organization of the client has enough skills and capacities to develop the CRM system?

Interviewer: Good question. They said they have enough people and capacity. Let's look at the organizational issue then. What do you think about the preference of the clients' IT people for the project?

Candidate: I figure they prefer in-house development because 1. They will get the cutting-edge skills of CRM system development. 2. Their voice in the organization would be stronger. 3. They are afraid of possible restructuring (layoff) when the CRM package is introduced – there will be smaller number of people needed.

Interviewer: Good. Now your client is considering introducing e-business in customer service management. For example, they plan to receive shipping orders through the web. What kinds of issues are needed to be analyzed?

Candidate: There should be a thorough cost/benefit analysis. For the cost issues, they need to calculate the total cost of the project and future variable cost changes. Also, they need to figure out how many of their customers are reluctant to use web channel.

Interviewer: Okay. What would be the measures for the benefit then?

Candidate: I think service lead time and service quality would be critical.

Interviewer: Good. What would be another benefit other than increased customer satisfaction?

Candidate: The client can understand the customer better through the analysis of the transaction data. They can analyze the customer behavior and spending patterns. Also, it would be easier to differentiate “most profitable customer groups” and drive more businesses from this customer group.

Interviewer: Very good! Now we’re running out of time. Thanks.

SELECTIVE BINDING CASE

Your client is a major fashion magazine that has been offered by its printer a proprietary new process called selective binding which enables publishers to customize the pages included in readers' magazines based on demographic data known about the reader. For example, an ad in Better Homes & Gardens for lawn chemical services could be placed only in those issues going to subscribers who live in houses and not to those living in condominiums or apartments. In this way, advertisers can focus their communications on the demographic segment they are targeting. Would you advise your client to take advantage of this new process and offer selective binding to its advertisers?

Analysis

This is a pretty straightforward cost/benefit analysis. The Magazine would want to consider offering the service to its advertisers if it would be able to enhance its earnings by being able to charge its advertisers a premium for being able to more exactly and efficiently target the demographic segment they want to reach. Of course the increased revenue from the any premium must be able to offset any revenue lost as advertisers stopped targeting. The interviewee could start the analysis by obtaining the following information from the interviewer:

- Q: What demographic breakdowns can be made in the magazine's database?
- A: The only breakdown possible on your database is between subscribers who make under \$50,000 and those who make over \$50,000.
- Q: What is total readership, the proportion of readers who are subscribers (as opposed to newsstand buyers), and the proportion of subscribers in each demographic category?
- A: There are 1 million readers, 80% of who are subscribers. Twenty-five percent of subscribers make under \$50,000, 75% make over \$50,000. The same mix applies to the newsstand buyers according to readership audits.
- Q: What proportion of the client's advertisers target each demographic category of readers?
- A: Most advertisers are selling high-end fashion products, so 75% of them are targeting the high income group.
- Q: What is the cost of the selective binding service and what does the magazine charge for its ads?
- A: The service is being offered to your client free for 3 years since the printer wants to promote the service's use by getting a major magazine to start using it. The client charges \$50 per thousand per full-page ad (selective binding can only be offered on full-page ads). Therefore revenue associated with a single inserted page (front and back) in an issue is \$100 per thousand.
- Q: What does the client's closest direct competitor for advertisers charge for ads and what is their readership like?

- A: The client's closest direct competitor has 500,000 readers, 100% of whom are subscribers. Effectively, all of their readers make over \$50,000. They charge \$70 per thousand for their full one page ads.

Since the printing cost to the client of selective binding is zero, the client simply needs to evaluate cost on the basis of revenue per thousand gained or lost as their advertiser base uses the service to better target their ads to their desired segment. Presumably, instead of 100% of advertisers paying the full \$50/thousand per page, the 25% of advertisers targeting the lower income segment will choose to advertise only to the 25% of subscribers targeting the high income segment will choose to advertise only to the 25% of subscribers falling into that segment and the 75% of the advertisers targeting the high income segment will advertise only to the high income subscribers (75% of subscribers). Assume that all advertisers continue to advertise in 100% of the newsstand copies. The revenue effect of this change can be calculated by looking at the impact the change would have on average ad rate per thousand on subscription readership:

New ad revenue per page = Old ad revenue per page X [(% low income subscribers X % low income target advertisers) + (96 high income subscribers X % high Income advertisers)]

Thus

New ad revenue per page = \$50 X [(25% X 25%) + (75% X 75%)]
at old rate \$31.25 < \$50

Now the question is, can ad rates per thousand on the selective binding portion of ads sold be increased sufficiently to increase average revenue per thousand over what it is today? To answer this question, your client's ad rates must be looked at from the perspective of their advertisers. If you consider the advertisers targeting the high income group, their alternative to advertising in your client's magazine is to put their ad dollars toward the 100% high-income readership competitor. The cost per thousand high-income readers with the competitor magazine is:

(Page rate X total readership)/ (portion of readers who are high income) = (\$70 X 500,000)/500,000 = \$70

Thus \$70 is the maximum price per thousand the client can charge its advertisers for selectively bound ads before the advertisers would switch to their competitor. Note that currently, the client is a cheaper buy for these high-income advertisers even though they are paying to reach readers they do not want:

(\$50 X 1 million)/750,000 = \$66.67

If the client charged \$70/thousand for selectively bound ads, average revenue per thousand to the client would be:

\$70 X [(25% X 25%) + (75% X 75%)] = \$43.75

Since \$43.75 is less than the \$50 that advertisers are currently paying, the magazine should not offer advertisers the selective binding service.

Of course, there are other issues which interviewees might want to mention such as the possibility of price discriminating between high and low income advertisers, the potential for and cost of expanding the advertising base using selective binding as a selling tool, etc. However, it is important by the end of the interview to have reached a recommendation regarding the initial question posed by the interviewer. To mention these other possibilities and areas for further investigation is certainly worthwhile, but it is also important not to get too far off track or to complicate the issue so much that a final recommendation is never reached.

SCIENTIFIC INDUSTRY

A manufacturer of scientific instruments is experiencing declining sales in its major product line. Why?

Approach

Here are some questions which may help isolate the key issues:

1. Describe the instrument and what it does. (Goal: gather background information on the product).
Response: The instrument, call it Y, is able to perform elemental mapping; that is, it is able to determine the specific composition of material placed in the chamber for observation. Y is an accessory for larger and much more expensive instrument that functions almost exactly like a microscope, which we'll call X.

2. What other products does our client manufacture? (Goal: gather background information on the client).

Response: They recently began manufacturing X, and also produce an unrelated product.

3. Can these instruments be used separately, and are they ever sold separately? (Goal: understand the sales process and the potentially interactive role of the X and Y sales forces).

Response: X can be used by itself, but Y is essentially dependent on X for its operation. As a result, except for replacement sales, Y is rarely sold individually. In fact, X's sales force will frequently recommend that a buyer purchase a certain Y while buying an X. Two years ago, over 30% of our clients sales were generated by a manufacturer of X.

4. What is the current %? (Goal: determine whether this could be a cause of the sales decline).

Response: It is currently around 5%

5. Does our product X compete with other manufacturers of X, and particularly the manufacturer that was selling our Y? (Goal: understand reasons for our friendly X manufacturer stopping promotion of our product).

Response: Yes it does compete directly with it, and our client introduced the product about 1 1/2 years ago. (You have discovered a significant portion of the sales decline).

6. How does our product compare to other Y's? (Goal: determine whether others are beating us on technological or other product features).

Response: Our client's product is regarded as one of the best in the market.

7. Is the market for X and Y growing, shrinking or flat? (Goal: a shrinking market could be a good explanation for declining company sales).

Response: Both markets are flat.

8. Who uses X and Y? (Goal: determine market segments).

Response: There are two basic user groups: industry, primarily semiconductor manufacturers, and academia (in research labs). What we've noticed lately is that the specific users in each of these groups, who also happen to be the primary buyers, have become relatively less sophisticated; that is, they are hired just to run the instruments and know less about their technical qualities. These buyers have become even more dependent on the sales forces. What has happened is that our client alienated itself from other manufacturers of X at a time when a strong relationship was becoming even more important than it used to be. The buyers are relying more and more on the X sales force, who is typically called well in advance of the Y sales force. (The interviewer will not likely give you all of this information at once. Questions about the buying process and changing decision makers would have brought it out)

This is the second part of the main reason for our clients declining sales: in addition to ruining our relationship with a manufacturer of X by producing our own, we happened to do so at a time when relationships became even more important.

PIPELINE COMPANY

You are hired by a large pipeline company to evaluate the current and future potential of the pipeline industry. The pipeline industry sprang up as transportation costs for mineral extraction companies began to escalate. There is currently 20,000 miles of pipeline throughout the U.S. What information would you want to know about the pipeline industry that could help you plot a strategy for a pipeline company?

Additional Details:

- **Industry Structure:** There are many pipeline competitors. Pipeline can be characterized as either common carrier pipelines (~70% of all pipeline miles) which are regulated by the government or proprietary pipelines (~30% of all pipeline miles) which are wholly located on the private property of a firm (e.g. a pipeline from a port station to a near-shore refinery). There are many suppliers of common carrier pipelines. The second group (proprietary) is not regulated by the government.
- **Products:** The pipelines carry liquid and gaseous materials -- crude oil, natural gas, methane gas, liquid nitrogen, refined oil products (gasoline), and chemicals.
- **Cost Structure:** There are exceptionally high fixed costs involved in a pipeline. The variable costs are primarily the electricity to power pumping stations along the pipeline. There are different cost structures depending on the type of product being moved. Pumping crude oil along the pipeline can cost as much as \$2M/month in electricity for a station. Gaseous products require considerably less energy to move.
- **Market Conditions:** U.S. proven reserves are diminishing and foreign imports are increasing. It is expected that for the next 5-10 years demand will be steady.

Key Points: (classic Porter analysis could be used -- This is rarely the case!!!)

- *Threat of Entry* is low because ...
 - there are high fixed costs (high initial investment)
 - pipeline services are essentially a commodity product (commodity markets are slow growth and unattractive)
- *Industry Rivalry* is strong because ...
 - there are many competitors and switching costs are low
 - industry growth is expected to be slow (i.e. market share is important)
 - many competitors use pipeline for in-house uses and only carry other products if capacity is underutilized
 - there are very high exit barriers (i.e. there is a strategic relationship between refining and piping)
- *Substitute Products* are many as witnessed ...
 - by proliferation of tanker cars and tractor trailer rigs for liquid and gaseous materials
- *Power of Suppliers* is not a significant factor.

Power of Buyers is not a significant factor because many pipelines are regulated and there are many buyers

- **Other considerations:**
 - **Product Mix:** The margins on gaseous products is higher than heavy unrefined products.
 - **Government Regulation:** Margins are greatly affected by common carrier status. Any future environmental regulations will cut even deeper into margins.

- Pipeline as a storage medium: For many firms the product in a pipeline can be a significant portion of its inventory and the volume in line must be considered in production. The classic question: Is it better to make product and sell it now at low prices or wait for prices to increase (e.g. crude oil prices)? A large pipeline could be a temporary storage facility.
- Operations: Maximizing profit means understanding the parameters of pumping -- costs of pumping at less than full capacity; layout of pipeline and pumping stations; products which can share the same pipeline; construction of parallel pipelines.
- Market Differences: The market for crude oil is very different than the market for specialty chemicals or natural gas. the pipeline manager must aware of these rapidly changing commodity markets to maximize his profit.

PACKAGING MATERIAL MANUFACTURER

Your client is the largest North American producer of a certain kind of bubble-pack packaging material. Currently, the company has 80% of the market, and has asked your firm to assess the strategic outlook for this company. How would you begin to assess the future for this client, and what type of recommendations could you make?

Information to be divulged gradually:

Costs for the product are broken down as follows: 20% for polyethylene, a plastic chemical. 35% conversion costs, including allocated fixed costs, labor and energy costs 10% distribution and storage, 15% marketing and overhead. Profit margins are 20%. Polyethylene is a commodity chemical. The factory is thirty years old, and the technology used is the same as when the factory opened.

The client had 100% of the market until two years ago. Since that time, a localized upstart company has appeared in the Philadelphia / New Jersey market and has captured nearly all of that market. This factory has purchased technology from a German company. Your client does not have much information about this competitor, but it appears that their factory is extremely efficient. They have also been undercutting your client on price.

Solution:

The competitor has used their new technology to produce a lower price product. As evidenced in the Philadelphia / New Jersey market, nearly all customers prefer this product to your client's. Therefore, the future is extremely bleak for your client, and they should be advised to respond to the competitive threat, perhaps by updating their own technology.

OVERSEAS CONSTRUCTION

An overseas construction firm wants to expand by establishing a presence in a growing U.S. regional market. How should it go about doing this? What factors are critical for its success?

Suggested framework:

What are the diversifying firm's distinct competitive advantages?

What is its capacity for funding an acquisition?

What is the competitive environment like in the proposed region?

How does this environment differ from the current markets of the diversifying firm?

Possible Solution:

Diversification could be effected through joint ventures or through acquisition. Which of these two strategies would prove the most suitable would depend on the availability of funds and on the nature of the companies operation in the region.

However, the success of the venture would depend not only upon the means of entry. Other critical factors would include:

- The existence of a distinct sustainable competitive advantage. For example:
- Non-unionized labor might help support a low cost production strategy (but for how long?)
- Proprietary technology not available to other companies in the region
- Special expertise in a growth area (such as, for example, hazardous waste)
- Access to distribution channels

OPERATIONS DIVISION

Your client is the operations division of a large company. The division is worth \$1 billion and is responsible for 50% of the operating expenses of the overall company. The perception within the company is that this division:

- 1. Costs too much**
- 2. No one knows exactly what they do**
- 3. The division should be outsourced**

You have been asked to produce a 5-year strategic plan for the company, and specifically to address the problems posed by this operations division

Information to be given when asked:

- The other divisions of the company are Investments and Insurance.
- The Operations division is responsible for accounting, documents, data warehousing, hosting services, technology and communication infrastructure like phones etc.
- The Operations division does a significant amount of back end processing for the other divisions, such as processing sales.
- The Operating division does not directly generate revenue, but the work they handle is essential for the company.
- There are no regular meetings between the representatives of the three divisions.

Suggested approach

- It should be clear that Operating division is a cost center and the goal should be to determine the worth of this division, and help the other two divisions realize this worth.
- Communication management is the key in this case, and this can help integrate the division into the larger company.
- The Operations division's employees should first be able to quantify the benefits of the work they do and explain it to the rest of the company.
- Suggestions may include working towards Activity based costing, where the cost bucket can be identified and associated with the other divisions.
- Formation of cross-functional teams could help all division better understand they needs and how best to meet those needs.
- May be able to identify the some standard tools/technologies/other backend functions, which can be outsourced to shave off some costs.
- Solutions should include an expected time frame, since the problem asks for a 5- year strategic plan.
- This case relates to improving the internal processes of a company and focusing too much on the company's products, customers or other such external factors is unproductive.

MAGNETIC TAPE MANUFACTURER – FACTORY RELOCATION ANALYSIS

Your client is a large manufacturer of magnetic tape rolls. They currently have one factory located in Omaha, NE, but are looking to improve their cost structure by relocating to either Mexico or Malaysia. Please analyze the cost drivers that would be affected by this move and determine which location is best. Please also discuss how you would implement this decision (assuming the client moves to one of the two locations).

Other information to be supplied if requested:

- The company currently produces 2,000 large roles and 5,000 small roles of tape per year.
- Factory size needed is 20,000 sq ft.
- Initial investment required to build a new plant is similar in both countries.
- Raw material costs are similar in both countries
- Client sells mainly to wholesalers in the US and Europe, but is considering entering the Asian market

	Mexico	Malaysia
Labor Costs	\$2/hr	\$1/hr
Labor Usage	• Large roles: 5 hrs • Small roles: 2 hrs	• Large roles: 5 hrs • Small roles: 2 hrs
Transportation costs	• Large roles: \$5/unit • Small roles: \$2.50/unit	• Large roles: \$10/unit • Small roles: \$5/unit
Rent	\$3/sq ft	\$2/sq ft
Total Cost	\$122,000	\$105,000

Suggested responses:

- According to the cost calculation, Malaysia is the better option. As the calculation is relatively simple, the interviewer should probe for deeper understanding of the non-financial issues.
- Candidate should discuss management challenges associated with moving operations overseas, supply chain/logistical issues, hiring a new workforce, and geopolitical risks.
- Other considerations: supplier/vendor relations, customer response, severance package for workers in Omaha, etc...
- Implementation plan should address logistical issues with bringing the new plant online and closing the old plant, as well as breaking the move into different work streams.

MACHINE-LOADING CASE

A client produces a range of synthetic materials in varying widths and lengths. Each material is used for packaging but differs in physical properties in terms of costs, weight, flexibility, and general performance. Each material can be coated with any one of four or five types of chemical coating which make the materials more or less impervious to heat, light, water, vapor, etc.

All of the machines on which these materials are made are housed in one enormous factory location. Each machine is capable of running any one of the various materials and/or coating combinations. The client does not wish to invest in additional equipment at this time.

The client has asked us what combination of products he should run to increase his plant's profitability. How would you go about determining the optimal mix of potential products on these machines?

If asked, you may provide the following information.

Tip for the interviewer: This is a macroeconomic case with a purpose to determine whether the candidate can dissect a general economic problem

Market Share

The industry is highly fragmented. A variety of small manufacturers supply similar products to provide a range of customers. Our client estimates he has less than 1 percent of the total market. No competitor has more than 3 percent of the total market

Cost

Each product has a different cost to manufacture dependent on materials used and the manufacturing process.

Price

Each product has a different price dependent on both the client's cost to manufacture as well as the market for the product.

Products

Our client's machinery can produce hundreds of different products. Some are unique to meet specific customer requirements while others are used by a wide variety of customer.

Customers

Our client's customers are primarily consumers or industrial product manufacturers who use the synthetic materials in packaging their own products.

Suppliers

Our client uses primarily commodity products in the manufacturing process. All can be obtained from a number of sources.

Note to the Interviewer

The primary issue of the case is to determine that the profit of the plant will be maximized when the most profitable product mix is produced and sold. The candidate should cover differences for each product in the fixed and variable manufacturing and selling cost and prices, as those must be determined to understand each product's profitability. The interviewee should also address the market demand for each product (to ensure what is produced can be sold at an acceptable price).

If the candidate is discussing issues which are not relevant to the profitability of each product line or to maximizing the profitability of the plant, repeat the question and ask how the issue being discussed will lead to a solution for the client.

Minimum Requirements

Candidate should, at a minimum, address the following issues:

1. Are there market limitations to the potential production of any one material?
2. Is there competition for these products?
3. Are there differences in costs in the manufacturing of these materials? For example, do some coatings cost more than others? Do some materials have inherent cost differences?
4. Is there flexibility in pricing of these products?

Additional observations should include:

1. Are there differences in setup time and cost for various materials or coatings?
2. Do these materials move at different speeds through the machines?
3. Are the machines truly interchangeable or are some better suited to one product or another?
4. Is there unlimited market demand for these products?
5. Are there technological displacement or replacement products on the horizon?

Outstanding Answers

The best candidates will formulate a profit maximization algorithm. The best algorithm is to maximize the profit contribution per machine hour.

1. Profit contribution is (unit volume) times (unit price minus variable cost).
2. Machine-hour capacity is a surrogate for fixed costs per unit of volume. Fixed costs take into account depreciation and standby costs as well as those costs that are independent of the variable costs per pound or ton produced.

An outstanding answer must include recognition of the asset costs and capital implied by that, as well as the income or profit contribution. Also, the potential substantial differences in volume produced per machine-hour and/or the price obtainable in the market demand and competitive actions.

LARGE BANK

A large bank has been around for 120 years. 30 years ago, they invested heavily on technology and automated majority of their transactions. However, with changing business conditions and the advent of new technology like wireless and internet, they realize they need an overhaul of their technology investments.

The CIO has commissioned you as a consultant to estimate the cost of porting their current IT infrastructure into a web based state of the art system. The ball park cost is to be estimated within 30 minutes. How would you approach this? In addition to cost, also come up with some key guiding principles of the estimation.

Solution

Me: We need to look at the problem from both business and technology standpoint. This is a good chance for us to re-engineer their business processes and their organization structure along with this IT investment. Should I look at that too? Also, should I look at both hardware and software cost?

Partner: No, just focus on Technology overhaul, and the software cost.

Me: Ok, we can take a bottom's up and top down approach to this. From a top down, we should benchmark the project against competitors who have already done this project.

From a bottom's up perspective, I would like to know about the lines of code in the solution, the cost associated with writing each line and hence the overall porting cost.

Partner: So, what do you want to know? Go with the bottom's up approach.

Me: How many lines were there and what is the cost of writing each line?

Partner: One million lines, and \$200 per day per developer.

Me: I assume each developer can write 50 lines. Using that, we can calculate,
 $1,000,000 / 50 = 20,000$ man days
Assume \$200 per man days, and we get, \$4,000,000 or 4 million.

Add to this management and business analyst overhead, 100%
And we get 8 million.

Partner: That's all you have?

Me: This is pure development, I would also add time for testing the software and deploying it, as 100%, so the total cost will be \$16 million.

Partner: Do you think this is alright? (Smiles)

Me: I think this is low, let me recheck my assumptions (I run through my calculations..) I think 50 lines per day is too high. I will take it as 25 lines per day. That will make it \$32 million.

Partner: Ok, what else? Any other factors to consider?

Me: Yes, this does not include hardware and any strategy consulting cost. Also, there are few things we should consider:

- Outsource or Do it yourself?
- Build versus Buy
- Phase wise implementation versus big bang
- Keep business team involved as part of IT team
- Design first and then code
- Use industry standard tools and practices and reuse available frameworks

Partner: You seem like a technologist

Me: I am from Tepper, the Mecca of business and technology

Partner: Thanks, and have a good day

INFORMATION SERVICES COMPANY

You are hired by a library information services company that provides a computerized article search product on CD-ROM. The product allows users in a library to locate articles by keyword search. The company currently has a weak market share of only 10% of all installed units. The company wants to understand (1) why they have so small a market share, (2) what could be done to improve the situation, and (3) where it should focus its resources.

Additional Details:

- **Competition:** There is a single major competitor which has 50% market share. The client and two other competitors each have 10%; and the remainder is divided among many competitors.
- **Market Segmentation:** The following table outlines many of the details of the market segmentation and client product data.

Type of Library	Number of Libraries	Client Market Share	Major Competitor Market Share	Competitive Features
Academic	5000	20%	60%	Search Quality, Content Content, Ease of Use Content, Ease of Use Price, Ease of Use
• Research	500	80%	10%	
• Other	4500	13%	66%	
Public	10000	10%	40%	
Secondary Schools	20000	~0%	10%	

- **Product:** The client sells a CD-ROM based product which is used on a dedicated PC in a library. The product has different versions that are upgraded each year. Each version is marketed to a specific library segment. Libraries are interested in matching the article search to hardboard volumes available within the library. The client's product is considered to have the highest quality of article search.
- **Pricing:** The client sells its product at a 25% discount to the major competitor and has the lowest prices in the industry. The pricing and profit schedule for each version are shown below.

Library	Client Price	Client Profit per Unit	Major Competitor Price
Academic	\$2000	>\$500	\$2667
Public	\$1500	\$500	\$2000
Secondary School	\$1000	\$100	\$1333

- **Competitive Features:** Competition within the industry focuses on four dimensions: (1) Search Quality, (2) Content, (3) Ease of Use, and (4) Price. The table above indicates the relative preference for these features for each market segment. There is a trade-off between ease of use and search quality. A better search requires a more skilled approach to keyword usage and often makes the search more difficult. The client's product is considered to have the highest quality search among the competitors.
- **Production:** the product is created by programmers who seek to match the product to library volumes. Since the principal input is labor, the type of CD-ROM created can be altered relatively easily.

Key Points

- The client's product does not match the needs of the large segments of the market (i.e. the client's high quality of search only appeals to a small segment of the total market) ==> weak market share
- The client should reallocate its resources to create products in the larger market segments -- products that emphasize content and ease of use over search quality.

The most profitable segment can be identified by using current client prices which should allow it to gain market share (due to the 25% discount to the major competitor) and calculating the maximum market profit. Academic = $5000 \times 500 = \$2.5\text{M}$; Public = $10000 \times 500 = \$5.0\text{M}$; Secondary = $20000 \times 100 = \$2.0\text{M}$. Therefore, if we realign our product to emphasize ease of use and content, the potential profit is $4500 \times 500 + 10000 \times 500 = 7.25\text{M}$ (minimum since profit in academic segment is $> \$500$ per unit).

FRENCH PIZZA MARKET

Pizza Hut has recently entered the home pizza delivery business in Paris. The market for home delivery is currently dominated by Spizza Pizza. Pizza Hut has asked your consulting firm to help it analyze issues that will determine its likelihood of success in the Parisian Pizza market. First, what information would you need and second, how would you analyze the pizza delivery market?

Possible Information Needs:

An estimate of the size of the Parisian home pizza delivery market would be useful. This could be obtained by knowing the population of Paris (6 million) and making some educated guesses about factors that determine pizza market size.

You may also want to know the size of Spizza, the current competitor, including sales, number of stores, and proportion of Paris that is currently served by Spizza.

Other useful information: market segments targeted and served by Spizza; market segments that are neglected by Spizza; what type of product do they offer; what do they charge for their product; what is the cost structure of their business and what products are most profitable.

Method of analysis:

The best method of analysis would start by determining if any part of the market is not well served currently by Spizza. Determine what the needs of any neglected market are, and understand if your client could profitably serve this market.

Also, try to understand the likely competitive response of Spizza to your client's entry. How will you defend your position if Spizza decides to fight for market share?

FINANCIAL SERVICES - MOVING

Your client is a financial services company that is moving its support operations from New York City to Des Moines Iowa. Please identify some of the primary benefits of this decision for your client, and also identify some areas of concern that your client should be wary of.

Interviewee:

Benefits: A company that does this may realize several monetary benefits. In particular, one would expect that a company should see:

- Reduced payroll expenses (lower salaried employees)
- Reduced tax expenses (lower tax rates), and
- Reduced rent expenses (lower lease payments).

Potential Issues: Any front office values its operations support highly, and moving the company's operations function may diminish the quality of these ops services. The lack of face-to-face interaction between the company's front-office and its support team can create communication breakdowns and significantly reduce service levels.

Service levels may also deteriorate based on the performances of inexperienced personnel. The company will likely have an extended period where hiring and training will need to occur in Des Moines, and the company should expect reduced operational effectiveness during this time (assuming the company's current operations employees will largely be unwilling to move from NYC to Des Moines and take a pay cut).

Other problems may include negative company morale, as front office employees will see their operations colleagues laid off. One might also be concerned that a different company culture may develop in Des Moines that would clash with the NYC company culture.

In summary, this client should expect to see a positive financial impact by moving its operations out of New York City. However, the company should be prepared to face some hard times as it creates distance between its front office and back office, and works through the challenges of building up its operations function remotely.

DRY CLEANING

Information to be given when asked:

- Market is more attractive in the urban areas, as compared to the rural because of the higher disposable income, and more need to dry cleaning service in urban areas.
- Existing competition is quite fragmented and no large chains dominate.
- Client's current brands of detergents are well established and there is a possibility of cross selling/promotional marketing campaigns
- Costs in establishing the new line of business will include fixed costs (land etc.), material, labor, and potential environmental costs
- Environmental issues are key. Dry cleaning requires the use and disposal of thousands of gallons of corrosive chemicals each year.
- Environmental issues make it a risky proposition. Costs to the client could increase because of potential lawsuits, expenses related to disposing corrosive chemicals, insurance etc.

Possible approach

- For part (i), Porter's 5 Forces framework can be used to determine market attractiveness.
- While considering the market entry, the candidate should also consider the role of regulations/environmental issues as one of the potential barriers to entry.
- Current client capabilities and how the client can leverage those to gain synergies in the dry cleaning business should be explored. Marketing/advertising is one potential area and the value of cross promotional marketing campaign should be considered
- Candidate should be able to identify various costs associated with the venture, the various revenue centers and compare the estimated cash flows for value add.
- Should be able to discuss ROI analysis using the NPV of dry cleaning facilities in major metros (first estimate the potential size of the market for dry cleaning in the US)

Should be able to explore the various risks associated with environmental issues and how these issues could make the cost of entry prohibitive

A.T. Kearney Round 1 Interview

Given at Carnegie Mellon's Tepper School of Business on 20 Jan 2006

Information Provided:

Our client is a prescription drugs distributor serving primarily retirement homes. In addition to 15 warehouses nationwide, the client operates about 200 local pharmacies. Doctors from the retirement homes fax prescriptions to the pharmacy, which then verifies the patient's information with the insurance company and issues a prescription. Fulfilled prescriptions are then loaded on a van and a delivery is made once a day to all of the retirement homes within that area. The transit time from the warehouse to the pharmacy is 3 days. A key differentiator of the customer is error-free prescription filling.

Question:

How can our client minimize costs?

Additional Info when asked:

- Pharmacies do not have retail opportunity
- Pharmacies are small offices served by 1-2 people
- Pharmacies are located in densely populated areas with a decent concentration of retirement homes
- There are some costs involved in setting up each new patient and verifying his/her insurance information

Additional questions:

What are some effective ways of inventory control?

Where should the warehouses and the pharmacies be located?

How can our client increase revenues by expanding their client base?

Suggested Answer:

This is a classic supply chain case. There are several parts to answering the first question. First of all it is necessary to understand that the demand for prescription medicine from retirement homes is pretty constant and can easily be forecasted. The predictability of demand simplifies ordering process and decreases the inventory on hand at the pharmacies.

Most of the inventory is stored at the warehouses. This model simplifies drug sorting as larger areas can be dedicated to the same drug.

The warehouses can be located anywhere where the real estate is cheap and labor costs are low (e.g. Campbellsville, KY). Pharmacies should be located in areas with high concentration of retirement homes.

Ways to increase revenue (think constant, predictable demand):

- Prisons
- Special care facilities
 - Cancer centers
 - Psychiatric institutions
 - Rehab facilities
- Hospitals are also a possibility, but the demand predictability becomes difficult.
 - People spend less time at the hospital, therefore client turnaround is quicker, which increases set up costs.

DISTILLED SPIRITS

You are consulting for a major United States producer of distilled spirits. Their primary products are a line of mid-priced vodkas and two brands of mid-range rum. Over the past few years, the business has become less and less profitable. What could be causing this?

Other information:

The split of product sold has consistently been 60% vodka / 40% rum over the past few years. The selling prices of the two lines are essentially the same. Overall sales are growing at about 3 to 5% per year, the same as the industry average for these product lines.

An analysis of the costs reveals the following:

- Production Costs have remained constant
- Advertising Costs have remained constant on average
- Distribution Costs have increased significantly

The products are sold throughout the country. In 27 states, where alcohol is sold in privately managed supermarkets and liquor stores, “open” states, shelf space is extremely expensive and trade promotions are critical. Such stores are also becoming less and less willing to hold inventory, which is increasing distribution costs by requiring more frequent deliveries. In the other 23 states, liquor is only sold through state regulated liquor stores. Distribution costs in these states is much lower, as there are far fewer outlets to service and central warehouses for the state-run stores. Advertising of alcohol is much more tightly regulated, and therefore, advertising spending is lower.

Solution:

A greater and greater share of the volume is being sold in the “open” states, with sales in these states increasing at about 10% per year. Sales in the regulated states are actually decreasing. Because the regulated states are less expensive to serve, and therefore, more profitable, the fact that they represent a shrinking portion of the total has caused total profits to decline.

A.T. Kearney Round 1 Interview

Given at Carnegie Mellon's Tepper School of Business on 20 Jan 2006

Information Provided:

Our client is a consumer packaged goods (CPG) company that sells cookies, crackers, donuts, and other snacks. The company is well established and has been around for 60-70 years. Annual revenues are \$20B and the company previously had a profit margin of 20%. Recently, however, that margin has fallen by 20%.

Company feels that this is the result of changing market trends. Older consumers that were previously customers are not buying the same snacks they used to eat for their children. There are also some spikes in materials costs. In particular, the costs of milk and sugar have risen. Previously sugar was purchased from outside the United States, but new tariffs have increased the import price.

What should the company do?

Additional Info when asked:

- Our client feels that the new trend toward health foods is the major cause and expected to be a long term shift.
- Our client's products are in nation-wide markets.
- Revenues have fallen across all product lines, does not appear to be a particular product.
- Client previously launched a health food product that failed miserably.
- Competitors are facing the same revenue and cost problems as far as we know.
- Three potential companies for acquisition:
 - Company A: \$300M to purchase, profit margin of 10%, growth of 5%, nationwide distribution network, 100% organic foods company
 - Company B: \$200M to purchase, profit margin of 20%, growth of 10%, not nationwide, but sells in about 80% of nation, has a diet plan, similar to south beach, also co-brands with a gym
 - Company C: \$100M to purchase, profit margin of 25%, growth of 20%, local brand, mostly on west coast, does have a large innovation house

Additional questions:

How would you rank the preferred acquisition of these companies?

Why? What are key advantages and disadvantages of each company?

Suggested Answer:

The key to cracking this case is analyzing the market and figuring out how the client can best regain control. Though the market is shifting towards health foods, which the client previously failed to enter, you need to think about why the client failed in this arena and how they could best go about successfully penetrating the market; i.e. new company acquisition.

$$\text{Profits} = \text{Revenue} - \text{Costs}$$

We have some data that revenue has fallen and some data that costs have increased. Work through each side of the equation.

Cost:

- consider purchasing in bulk or establishing long term contracts to decrease cost of milk and sugar inputs
- consider buying milk internationally (price increase given was only domestic)

Revenue:

- look at changing branding (perhaps people don't want to buy health foods from a junk food company, could this be why previous health product failed?)
- consider changing marketing
 - create a new healthier look to older snacks, new packaging
 - market to children instead of parents, then they'll encourage parents to buy
 - consider marketing more internationally
- consider developing/producing a new health food
 - produce same products in low-sugar, low-carb, low-calorie, low-fat versions
- consider acquiring health food companies
- seems that junk snack foods and healthy snack foods are somewhat similar, so not a gigantic leap into new market. Company should be able to leverage current production facilities, distribution networks, industry expertise

CONSULTING FIRM STRATEGY

You are the newest member on the management committee of a well-known top-tier strategy management consulting firm. Eager to be accepted by your more senior peers, you volunteer to study the industry and propose a firm strategy for the 1990's, which you will present to the committee at its next meeting. As you leave the meeting you begin to realize the enormous task to which you've committed yourself.

- 1. How do you evaluate the consulting environment and determine likely future scenarios?**
- 2. What information do you use in this process? How is this information obtained?**
- 3. What do you believe is most likely to happen in the consulting industry given your present knowledge? How did you arrive at this conclusion?**
- 4. What strategy do you propose to the management committee?**

Proposed Answer

This is one of the most difficult types of cases because the answers are completely unknown and will vary substantially depending upon the interviewee's knowledge of the industry. This is also an interesting case since the salience is likely to be high. As an interviewer you should feel free to add information on an as-needed basis. When information isn't available, ask the Interviewee to develop his or her own hypotheses. What matters here is the thinking process, not necessarily the answer.

1. A good place to begin is to evaluate the industry from a competitive analysis perspective, such as Porter's five forces. The following is an abbreviated analysis.

Rivalry (low to moderate): management consulting is fragmented, with many players each holding relatively small concentration of total market. Firms act as competitive monopolists, and differentiate themselves by specialty, type of customer (Fortune 100 versus Fortune 1000 companies), reputation (McKinsey versus accounting firms), and the resources they employ (top MBAs versus all MBAs). Many companies are relationship-driven with their customers, which limits competition and keeps prices high. Top tier firms in particular are able to have high price points.

Potential Entry (moderate): there are no great barriers to entry into consulting; however, few new consulting firms truly compete in the top tier. It's possible new firms would enter if the industry were earning positive economic profits and if they faced certain imitability (e.g. the ability to recreate what the top tier firms do).

Substitutes (moderate): companies can move the consulting process in-house by hiring ex-consultants and bright MBAs.

Buyer Bargaining Power (moderate-high): In the last decade the consulting market has boomed, with supply generally following demand, which lowers buyer power. However, it is appropriate to question effect recession might have on industry. It's possible that demand may decrease as companies quit expanding, which would reduce demand, give buyers more bargaining power, and push prices lower.

Supplier Bargaining Power (low-moderate): Major suppliers are the intellectual capital employed by firm (e.g. experienced consultants who bring in sales and new consultants who provide analytics). Must pay market price or risk losing suppliers.

Other interesting points might explore the key success factors in the consulting industry. What sets top tier firms from middle ones? Do any firms have specific sustainable competitive advantages? How does the marketing mix differ among firms? Does your firm have any specific core competencies or advantages that set it apart from other companies?

Determining likely future scenarios is more ambiguous. There are at least several key point: what effect will a recession have on consulting firms? Will top tier firms suffer differently from others? How will the mix of products demanded change (e.g. cost-cutting studies rather than market expansion studies)? Will the consulting market continue to expand or suffer a cutback? Or, will certain geographical areas expand (Pacific Rim, Eastern Europe) faster than others? Again, the thought process is more important here than actual answers.

2. Information gathering is a key reason companies use consultants. An interviewee should have a decent understanding of business information sources and how information is gathered.

Information can be broken into two groups: secondary and primary. Usually one begins with secondary material, specifically, a complete review of published literature (a "lit search") pertaining to the study (e.g. journal and newspaper articles, investment bank research, specialized studies, books, etc.). This often points towards other good sources (e.g. industry experts, associations, major competitors, government sources, etc.). Hypotheses are often created from the secondary information. Primary research is then used to focus in on the key issues. This research includes telephone interviews, in-person interviews, mailed questionnaires, focus groups, laboratory experiments, etc.

3. This answer will depend upon the material covered in the first two. Ask the questions: What trends are likely? What is a positive scenario? A negative one? If you had any information at your disposal, how could you get a better handle on this issue?
4. There is no right answer here, so the interviewee may balk. However, you can provide some structure. What are the key success factors to succeeding in the industry? Is there any way to achieve sustainable advantage which cannot be duplicated by your competitors? Can you use non-traditional methods to achieve competitive advantage, such as leveraging through technology. Given your firm's competitive strengths and core competencies, what is the best strategic route?

CONSULTING FIRM

You are the managing director in a large international consulting firm. Traditional strengths of your firm have been solving strategy and organizational issues. Recently, you have noticed an increasing number of your firm's proposals are being rejected because of a lack of information technology expertise in your firm. So far, your firm's growth has been strong enough that proposals lost have not hurt annual earnings. Nonetheless, you are becoming increasingly concerned about the need to develop the firm's capabilities in information technology.

Q1: Assuming your concern is valid, what reasons will you provide to other partners about the need to acquire information technology skills?

Q2: Assuming you are able to convince other partners of the importance of IT expertise, what steps would you take to rapidly build IT capacity in this area?

Q3: What are the major risks in executing an IT capacity-expansion?

A1: Good answers focus on the value of IT to clients: discussion topics include the increasing importance of information in business, strategic value of information and information flows, importance of information systems for implementing new organizational structures and management control systems. Better answers focus on the costs of losing clients to competitors: discussions included the encroachment costs of having clients talking with competitors about IT problems, risk of losing credibility with clients by not being able to solve a problem.

A2: Good answers will focus on various methods to build expertise: buying expertise by acquiring another firm, by raiding IT practices of other firms for a few key consultants, building capacity through recruitment of IT experts and training them to be consultants, building capacity by training current consultants in IT practice skills, establishing a strategic alliance with a IT boutique firm. Candidates should discuss the pros and cons of each method proposed; impact on firm's current culture, cost to the firm, time needed to build expertise, etc. Better answers will realize the importance of stimulating client demand as capacity builds through seminars, articles strategic studies in IT areas...

A3: Good answers depend on the expansion methods discussed, but an important issue is the loss of the firm's focus away from just strategy and organization. Better answers will focus on the difficulty of implementation in IT; rapid technological changes in the IT industry require significant ongoing training and development costs; new practice cultures may be significantly different from current culture, especially if "external experts" are brought into the organization.

CONCRETE MANUFACTURER

Your client, a concrete manufacturer is considering acquiring a small local firm. What factors should be considered? After considering these factors, would you recommend the acquisition?

Suggested frameworks:

This is an acquisition question; look at company strategy, and revenue-cost synergy. Then focus on combined capabilities, culture similarities, competitive response, and financing. Also, consider any external impact that may be relevant, such as company image, etc.

Additional Information to be divulged gradually:

The target firm is currently profitable, with margins of 5%. Your client's margin is 15%. Your client attributes its higher profit margin to economies of scale in trucking and mixing, and a stable labor force.

Both companies compete in the geographical market, the Southeastern U.S. Your client's customers are large construction firms and contractors generally in the office and commercial building construction business. The smaller firm sells mainly to other small businesses and contractors. (Swimming pool installation firms, patio builders, etc.)

Additional research shows that the smaller customers for concrete are growing, while the major office building construction market is stagnant. The smaller firm has strong contacts with many local customers, and is often the preferred supplier due to their customer responsiveness.

Your client is not able to fund the acquisition internally, but could obtain bank financing at a rate of 10%. Similar acquisitions generally are made for two to three times current sales of the target firm.

Solution:

From a financial point of view, the acquisition is not attractive if there are no synergies between the firms. With profit margins of only 5%, the income generated by the smaller firm will not cover the capital charges (interest due to the bank) on the acquisition price. (Acquisition price = 3 x sales. Interest on this amount will be 10% x 3 x sales, or 30% of annual sales. Profits are only 5% of sales. This analysis, of course, ignores the tax shields.)

However, if your client were able to use some of its competitive advantages to improve the financial outlook of the target firm, the acquisition would be advisable. It is reasonable to expect that synergies would arise from economies of scale in trucking and mixing, which could raise the profit level of the target firm, and make the acquisition more attractive.

CHILLED BEVERAGES

You are consulting for the manager of a division of a large consumer products company. Her division produces fruit juices in three forms, all marketed under the same name: chilled (found in the milk section of the supermarket, usually), juice boxes, and frozen concentrate. This division has sales of \$600 million per year. The entire company has sales of over \$20 billion. The chilled segment represents \$120 million in sales per year. While juice boxes and frozen concentrate are profitable, chilled juices are only breaking even in good quarters and losing money in bad quarters. She has received a proposal from upper management to sell the chilled juices business. What would you advise that she do?

To be divulged gradually:

Chilled beverages is a \$5 billion dollar industry nationwide. There are two large players that have 40% and 25% of the market, respectively. Your client's market share, 12%, makes her third in the industry.

The best available information indicates that the two market leaders are profitable. The two market leaders are able to fund more advertising and more promotion, trade and coupons than your client.

The market leaders produce pure orange juice and blends that are based on citrus juices. Your product uses more elaborate blends of juices, usually with a base of pear or peach juice (95% of the inputs) and flavored with cranberries, bananas, mangoes, etc. (the other 5% of the inputs). Pear and peach juice are about the same price as orange juice, but the other flavorings cost about twice as much.

The market for chilled juices is essentially mothers with school age children. This is a highly price sensitive market that loves coupons, promotions, etc.

Brand name is important in this market, as in juice boxes and frozen concentrate, as mothers tend to prefer highly reliable products for their children. However, the brand premium must be in line with other branded products. Therefore, all branded juices tend to sell in the same price range.

One plant in California produces all of the products, chilled, juice boxes and frozen. It would be difficult to find another use for the plant without a major conversion.

Solution:

There are three choices:

Sell the chilled juice business. This would, however, affect the juice and frozen concentrate businesses, as there are both advertising and manufacturing synergies.

Sell all of the juice business. This may be more feasible, as the buyer could capture the synergies, but would not be too likely to turn the business around. The selling price is likely to be low.

Keep the chilled juice business and rework the ingredients and costs. This turns out to be the most feasible option, as evidenced by the success of the competitors.

MERGER CANDIDATE IN CHEMICAL INDUSTRY

One major chemical producer has retained McKinsey to evaluate another major participant in the industry. Both companies are bulk commodity chemical producers. We have been asked to begin our work by analyzing the future prospects of the target company's major product line, a bulk chemical used in the production of plastics.

Essential facts included:

- Production of this chemical has slowly declined over the last five years
- Prices have declined rapidly
- There are 7 to 8 major producers; the largest producer has a 30 percent share; number two has 20 percent; out target company has 15 percent; the rest is divided among other competitors
- The two largest competitors earn a small return; target company is probably at break-even; rest are operating at break-even or loss
- The largest competitor has just announced construction plans for a major new plant.

QUESTION

How would you structure an analysis of the target company's future prospects in this product line?

MINIMUM REQUIREMENTS

The candidate should, at a minimum, address the following issues:

1. What markets use this chemical, and what has been the nature of growth in these markets? (End-use markets are largely automotive-related.)
2. How much overall capacity exists now? (Far too much.)
3. What has been relative capacity utilization of competitors in the industry? (60 to 70 percent for last 3 years).
4. What are relative cost positions of competitors? (related to size/efficiency age of plant; target company has reasonably "good" position.)

BETTER ANSWERS

Better answers will move beyond the previous answers to consider:

1. How rational is pricing? (Prone to self-destructive cuts to gain temporary share points.)
2. Are there niche or value-added uses for chemical? (Not really.)
3. Does the chemical have a major by-product or is it a by-product? (Not of significance.)
4. How often have companies entered/exited, and how expensive is entry/exit? (Entry expensive; exit cheap for most because older plants are fully depreciated.)
5. How important is this product line to each of the competitors? (Most producers are diversified.)

OUTSTANDING ANSWERS

The best answers could address:

1. Reasons for announced capacity expansion. (It is a bluff to try and get smaller competitors to shut down.)
2. Is regulation important? (Yes: all competitors have installed pollution control equipment.)

3. What is nature of operational improvements that target company could make? (lots.)
 4. How is product sold and distributed? (Economies of scale in marketing and transport are critical.)
- Is there synergy between our client and target? (not really.)

CABLE TELEVISION COMPANY

Your client is a small holding company that owns three cable television companies in the Northeast: Rochester, NY, Philadelphia and Stamford, CT. Each of these three companies is profitable, and each has been experiencing steadily growing sales over the past few years. However, the management feels that the Northeast is not the fastest growing area of the country, and, therefore, acquired another cable television company in Tucson, Arizona a little over a year ago. Despite every effort of management, the Tucson company's sales have been stagnant, and the company has been losing money. How would you analyze this situation, and what could be the cause of the poor performance of the Tucson cable company?

To be divulged gradually:

The Tucson area is smaller than Philadelphia, but larger than Rochester and Stamford. Tucson is also growing at 12% per year on average. Per capita income is higher than in Philadelphia and the same as in Rochester and in Stamford.

Operating costs in Tucson are essentially the same as in the other markets. The cost of programming is based on number of subscribers and is equal across the nation. Operating costs are composed of variable items: sales staff, maintenance, administration and marketing. Only maintenance is higher than in the other markets, due to the larger land area serviced. Fixed costs relate to the cable lines, which is a function of physical area covered.

The Tucson company has attempted marketing efforts in the past, such as free Disney programming for one month, free HBO for one month, free hookup, etc. These programs have been modeled after the other three markets.

Cable penetration rates in the three Northeastern markets average 45%. The penetration rate in Tucson is 20%. These rates have been steady over the past three years in the Northeast. The penetration rate in Tucson has only risen by 2% in the past three years in Tucson.

There is only one real substitute good for cable television: satellite dishes. However, many communities are enacting legislation that limits their usage in Tucson. They are also prohibitively expensive for most people.

Solution:

The real error of management results from their failure to recognize another "substitute" good: no cable television at all; television reception is far better in the desert Southwest than in Northeastern cities. The lower penetration rate is most likely a result of different climate conditions and lower interference in Arizona.

BANK OF LUKE

Mr. Check is the Director of Retail Lock Box Services for the Bank of Luke, a medium-sized Midwestern bank. The Retail Lock Box Department consists of 100 clerks and 8 managers and supervisors. Each year, in addition to their handling of retail lock box transactions, the Department generated \$1.5 million of fee revenue processing retail credit card and mortgage payments ("items") for 75 commercial accounts. The bank has many other commercial accounts that use other companies of their item processing. In fact, the Bank recently lost the item processing business for one of its largest accounts to Vader Inc., the largest item processor in the US

The item processing industry has undergone dramatic changes in recent years. Types of items processed include credit card, mortgage, and utility payments (checks), airline tickets, and coupons. In the past, these items were usually processed by the issuing company (e.g., airlines would process their own tickets) or by bank item processing departments like the Bank of Luke's. At banks, the processing of payment items was done more as a service to bank customers rather than as a profit-making endeavor. Hence, it received little focus from management. Historically, processing was accomplished by verifying the correctness of incoming paperwork and manually sorting, filing, and totaling the items: only the largest banks were highly automated.

Companies specializing in item processing have emerged in the past ten years. Vader, Inc., the largest such company, is a subsidiary of a small bank in Georgia. Each year Vader processes millions of airline tickets and retail payments for hundreds of companies, most of whom are not customers of its hundreds of competitors most of whom are not, customers of its parent bank. Vader uses high-speed processing equipment and is highly automated. Processing time is rapid and processing costs are low. In fact, because of this speed advantage, the parent bank is beginning to profit from the float of checks processed. Although industry wide a majority of items are still processed by the issuing company or by small processors, it is expected that large processors. Within five years, it is expected that most of the business will continue to migrate to Vader and other large processors. Within five years, it is expected that Vader and the large processors will dominate this market.

Vader had a significant cost advantage over smaller operations, such as the Bank of Luke, because of the great economies of scale they gain from processing such volumes of items. In addition, Vader benefits from a more constant workload by processing both airline tickets and retail lock box receipts: airline tickets have few peaks and valleys, whereas mortgage payments always peak early in the month with very low volumes the rest of the month. Mr. Check believes that Vader quotes prices of 20 cents per item to large prospective customers while the Bank of Luke processes items for 40 cents per item.

The President of the Bank, Mr. Kenobi, has asked Mr. Check to evaluate how the retail lock box service can be made profitable; the service lost \$100,000 last year. Mr. Check believes that the bank must offer retail lock box services, and it must price the service to be competitive with companies such as Vader. Recognizing that outside expertise will be needed, the President has given Mr. Check a budget to be used to hire a consulting firm. Mr. Check has asked you to visit his office to discuss the proposed engagement. While walking to his office, you observe that the Bank's retail lock box operations remains primarily a manual system, with limited use of modern, high-speed equipment and methods. Once in Mr. Check's office, you note a picture showing the Department's staff in 1965; Mr. Check was a supervising clerks at that time. After reviewing some background information with you, Mr. Check asks you the following questions:

Question #1

What do you see as your (the consultant's) role at the Bank of Luke?

Question #2

What steps would you take and what information would you gather to diagnose the problems facing the Retail Lock Box Department and to develop solutions to those problems?

Questions #3

From what you now know, what are the problems facing the item processing service and what recommendations would have the greatest impact on the performance of the Bank of Luke and the item processing service?

Answer

In this case, we want to test the candidate's ability to handle a case in which the events appear hopeless until the end. When an apparently easy solution (automation) is made available. The candidates should challenge the general premise of the case, and not simply believe that the business is necessary just because Mr. Check says so. We also want to test creativity with this case. We purposely leave the case rather vague, not suggesting any particular actions and offering little data. The candidate should be given time to think about this case and propose solutions which are not readily apparent:

- Why not sell the business of these customers?
- Why not offer increased services to justify higher fees?
- What is the strategic plan for the bank, and how does this unit fit into it?
- What does Mr. Check feel his unit should be generating? (after all, \$15,000 per employee is pretty low!)
- Has he considered acquiring other banks' customers to increase the economies of scale in his own operation?

This case can also be used to discuss cost-cutting. Again, creativity and sensitivity to the real issues should be the goals of your probe; cutting 25% of the staff is too obvious and too easy.

A.T. Kearney Round 1 Interview

Given at Carnegie Mellon's Tepper School of Business on 20 Jan 2006

Information Provided:

Our client is a major automaker in Detroit. In constant efforts to eliminate costs, the client decided to implement a Just In Time (JIT) delivery of parts and components from parts manufacturers straight to the assembly line. This method eliminates costs of parts storage and provides greater efficiency for the production. The JIT method in this particular case works as follows: as a red-color car travels through the assembly line, parts specific for this color (interior, dashboard, etc) are automatically dispensed. Same principle works for any sequence of cars on the conveyor belt e.g.: Red Green Green Blue Blue Blue Black.

The success of the JIT system depends on vendors delivering parts in the correct order. Currently the vendors are charging our client a premium for delivery of parts in such a way. You've been hired to assess whether the extra charges imposed for the JIT delivery are feasible as well as determine what should be the key drivers for these costs.

Additional Info when asked:

- Parts vendors set costs based on a percentage of the total cost of the part.
- No similar models have been done before
- Assume that data is available

Additional questions:

Once the information is assembled, what's the appropriate method to determine the key driving factors

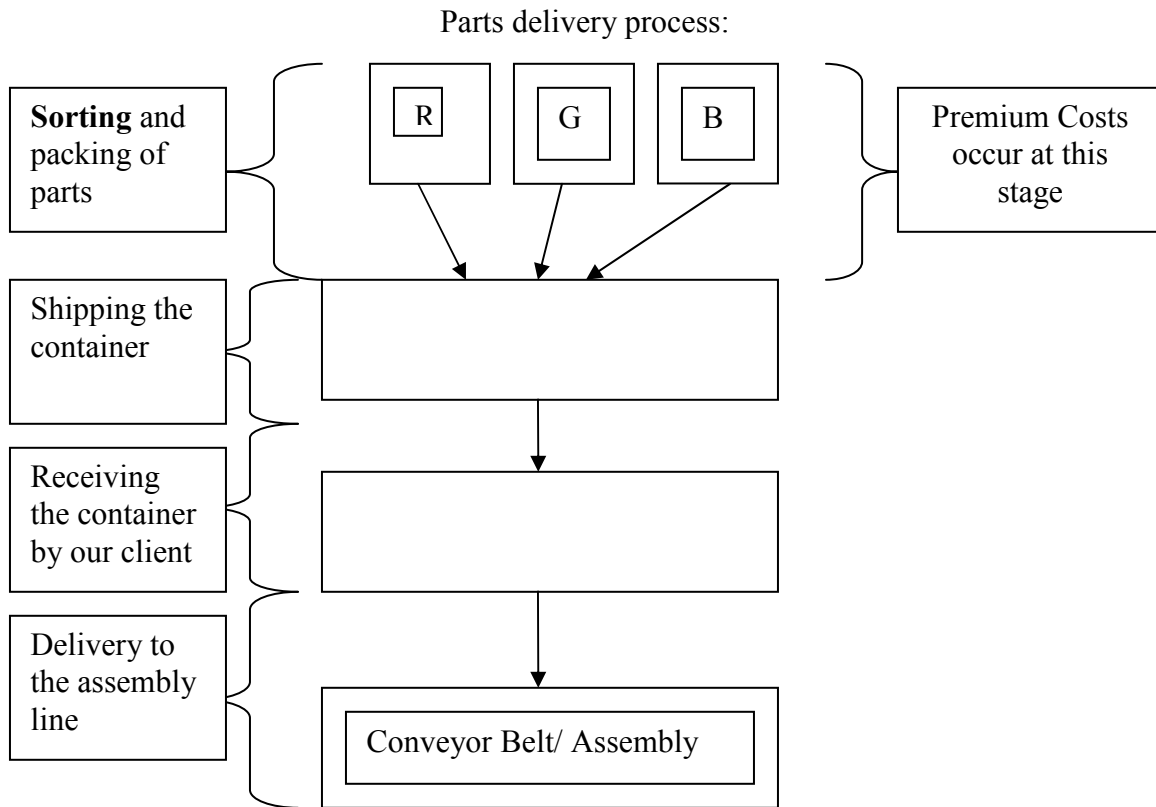
Suggested Answer:

Once the process of parts supply is split into the four parts, (see below) it is evident that the majority of costs occur at the sorting stage. These costs can be further split into the following:

- Sorting Machinery cost (depreciated over 5 -10 years)
- Labor – this is a major part because of unions and governmental regulations - for JIT the sorting process should work 24x7 -> overtime.
- Space – extra area
- Electricity
- Special packaging
- Weight/Bulkiness of the part (seats vs radio knobs)

After major factors have been identified, the appropriate method to find out key driving factors is to assemble a regression model. It will not necessarily be linear, so Ramsey tests need to be done. Don't forget to mention the White's AR's and heteroskedasticity

(?) tests. Describe the Chow Break-point test for extra credit. Also don't forget to mention that data for analysis is rarely clean and will require extra efforts to transform into a usable form.



AUTO MANUFACTURER

Your team is hired by a large U.S. automobile manufacturer (GM). They are interested in your evaluation of their \$10B after-market parts business. This business can be segmented into two sets of buyers: dealers authorized to sell GM parts (\$8B) and non-dealer merchandisers (\$2B). This second group can be subdivided into mass merchandisers and “service” providers. Mass merchandisers are of two types -- those which specialize in auto parts (e.g. Auto Zone) and those which sell diverse products including auto parts (e.g. Sears). “Service” providers include Goodyear or Western Auto.

GM would like for you to answer two questions: (1) Is there an opportunity to expand this part of the business? (2) How would they go about doing it if they chose to expand?

Additional Details:

- Company Economics: There are tremendous fixed costs in the auto business (including labor). All of GM’s parts manufacturing facilities are fully depreciated and they currently have excess capacity.
- Competitors: While Ford and Chrysler make parts for their own cars, they are not nearly as integrated as GM and tend to focus in specific parts categories. There are hundreds of small parts manufacturers which tend to focus on commodity-like auto parts (e.g. oil filters).
- Products: GM produces a full spectrum of parts classified as either *platform-specific* or *universal*.

	Platform-specific Parts	Universal Parts
Types of Parts	Body panels, brakes, transmissions, engines	Spark plugs, filters, hoses, batteries
Market Characteristics	Sold through dealers under warranty; high margins/low volume	Sold through many outlets; high turnover; strong competition; slim margins/very high volume
GM Sales	\$8B	\$2B

- Growth Rates: The table below provides the basic facts about each market segment’s growth rate.

Market Segment	Overall Market Growth Rate	Total Market Size
Dealer-authorized	-35% per annum	\$40B
Non-dealer		
• Mass merchandisers	+65% per annum	\$70B
• Service providers	+15% per annum	\$30B

Key Points: (Porter Five Forces analysis, *remember this is just one of many good approaches*)

- Threat of Entry is minimal for a broad category because the fixed costs are very high. However, a manufacturer could go after a niche play if it were to develop an advantaged cost structure or superior product. Switching costs among consumers is very low.
- Industry Rivalry is important for the mass merchandiser category because margins are slim (meaning price wars are more prevalent). Brand names (e.g. Fram, AC Delco, AutoLite) are important to many consumers.
- Substitute Products are relevant only in the sense that there are many competing products and future technologies such as electric cars could eliminate the need for many types of parts.

- Power of Suppliers is not a significant factor because inputs are commodity raw metal and rubber.
- Power of Buyers is important since there are few mass merchandisers such as Sears or Kmart and they demand full range of products and tremendous volume discounts.

GM's Position: GM may have a cost advantage due to its fully depreciated plants and excess capacity in a fixed-cost environment. Thus its variable costs must be below sales revenue. Also, its brand names are respected and are valuable to merchandisers in maintaining margins. GM's ability to produce a full-range of products is also an advantage. These advantages combined with the high growth rates for the non-dealer merchandisers should motivate GM to expand its business in this segment. GM should use its cost advantage, brand names, and full range of products to go after the most lucrative market -- the mass merchandisers.

ALUMINUM CAN MANUFACTURER

An aluminum can manufacturer has discovered a way to improve its manufacturing process. As a result, its manufacturing cost has been reduced from \$0.89 to \$0.79 cents. How can the manufacturer best exploit this cost advantage?

Suggested frameworks:

Remember basic economics. The firm can either use a penetration strategy or price skimming strategy. Consider the impact of either strategy on the company and its competitors. Also, don't forget to think about any substitutes for aluminum cans.

Interviewer Notes:

Clearly, the client should either drop price or reap additional profits.

It turns out that the client is the leader in its market with a 40% share and supplies directly to major beverage manufacturers. The number two player in the market has about 30% of the market and the rest is shared by many small competitors.

Aluminum cans have a lower priced substitute, steel cans, which have inferior printing and stamping characteristics. Steel cans are used by customers who do not want to pay the premium for aluminum cans.

If the client drops prices, other competitors will have to follow since this is a commodity market and not following would mean a quick demise. The lowering of prices might increase the client's market share marginally, but some smaller competitors will have to start exiting the industry and larger competitors will have to start investing to discover the client's cost advantage.

At the same time, steel can users will start switching to aluminum cans, thus hurting manufacturers in that market. The resulting growth in the aluminum can market will attract steel can manufacturers to enter it. Since some steel can manufacturers have deep pockets and a strong backing, these new entrants could pose a future threat to our client. In conclusion, it is best to retain prices and generate extra profits for now. The cost advantage may help another day during a price war.

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AGRICULTURAL EQUIPMENT MANUFACTURING

Your client is a large agricultural equipment manufacturer. Their primary product line, farming tractors, is losing money. What questions would you ask of your client to help them solve their profitability problem?

Answer: Agricultural Equipment Manufacturer

A: It is unlikely that there are too many players in this market. You might want to start off by asking how many competitors there are. Suppose the answer is that there are two direct competitors.

What is your client's market share relative to their competitors (your client has 40% of the market, competitor #1: 30%, competitor #2: 15%, with the remaining 15% belonging to many small manufacturers.)

What are the market share trends in the industry? (Five years ago, your client had 60% of the market, competitor #1, 15%, and competitor #2, 10%. Obviously, your client has lost significant market share to its two competitors over the last few years.)

Do all three competitors sell to the same customers? (Yes)

How is your product priced relative to your competitors? (Your client's product is priced higher than the others.)

Has this always been the case? (Yes)

Are the products the same? (Essentially yes, they all have the same basic features. Of course, tractors are not commodity items and a few differences do exist.)

What are the differences that allow you to charge a premium for your product? (Your client has a strong reputation/image of quality in the market and the market has always been willing to pay a premium for that reputation because it meant they would last longer and need less maintenance. This can be critical for some farmers because they cannot afford to have a piece of equipment break down at a critical time.)

Are sales revenues down? Are sales quantities down? (Yes)

Is the price down? All costs the same? (No, in fact both the price and costs are up.)

Have fixed costs increased? (No, material costs, (variable costs,) have gone up out of sight, and the client has no answer as to why material prices have gone up so staggeringly.)

Do you manufacture your tractor or just assemble it? (Primarily an assembly operation.) Finished part prices have gone up? (Yes)

Raw material prices for your suppliers? (I don't believe so)

Have labor costs increased for your supplier? (No)

Have you changed suppliers? (No)

Why are your suppliers charging you higher prices for the same products? (Well, they're not, the prices have increased as a result of our product improvement efforts. We've tightened tolerances and improved the durability of our component parts.)

Why do you make these improvements? (Because we strive to continue to sell the best tractors in the world.)

Are your customers willing to pay for these product improvements? (What do you mean.)

Are your customers willing to pay a marginal price which will cover your cost of implementing these improvements? (I don't know, I guess we assume that they will...)

It turns out that prices have been raised to cover the costs of these improvements, but customers do not value these improvements unless they are essentially free --so sales are down. The client needs to incorporate a cost/benefit analysis procedure into its product improvement process. Don't forget though, that you must consider the long-term effects of these decisions.

VIDEO GAMES

The CEO of a large diversified entertainment corporation has asked a McKinsey team to examine the operations of a subsidiary of his corporation that manufactures video games. Specifically, he needed to know it. He should approve a \$200 million capital request for tripling the division's capacity.

You are a member of the McKinsey team assigned to this project. Assume you and I are at the first team meeting. What are the critical issues we should plan to examine to determine if the industry is an attractive one for continued investment and why?

OTHER INTERVIEWER INFORMATION

The following information may be given if requested by the candidates though you should focus on having the candidate identify issues, not obtain more information.

Market share

Division is third largest manufacturer of hardware in the industry with 10 percent market share. Top two producers have 40 and 35 percent market share. Remainder is divided by small producers. Division sells to broad range of consumers.

Sales

- Division sales have increased rapidly over last year from a relatively small base. Current estimate is annual sales of 500,000 units.
- Current estimate of industry hardware sales is 5,000,000 units annually. Industry growth has been strong though over last few months, sales growth has slowed.
- Divisions current sales price for the basic unit is \$45 per unit.
- Division remains less than 20 percent of parent company sales.
- Top two competitors also develop, manufacture and sell software/games though division sells only licensed, software.
- Industry growth of software continues to increase.

Costs

- Division estimates current cost is \$30 fully loaded. Requested expansion should reduce the cost by 5 to 7 percent and triple production of the hardware units.
- Top two computers are estimated to have a 10 to 15 percent cost advantage currently.
- Main costs are assembly components and labor.

Customers

- Division estimates much of initial target market (young families) has now purchased the video game hardware.
- No large new user segments have been identified.

Distribution

Primarily outlets of distribution are top end electronics stores.

Profitability

Division currently exceeds corporate return requirements; however, margins have recently been falling.

Product

Hardware standards have been established by the industry leaders.

Product features constantly developed (e.g., new remote joy stick), to appeal to market segments.

Note to the Interviewer

The primary issue of the case is to determine if the industry is attractive and, especially, if our client's position in that industry is sustainable. The candidate should identify issues which are necessary for assessing both the industry and our client's position, but should not be expected to solve the problem.

If the candidate begins to discuss too deeply a specific issue, before having covered the key issues overall: bring them back to discuss the Industry more broadly by asking "what other issues must be examined?"

If the candidate is discussing issues which seem irrelevant to the attractiveness of the industry, ask "how will that analysis help to assess the attractiveness of the industry or our client's position?". Then, ask the candidate to identify other issues which must be examined.

MINIMUM REQUIREMENTS

The following issues would need to be covered for the candidate to have done an acceptable job:

1. What is future market potential? Candidate needs to question the continuation of overall industry growth. She/he might ask about the saturation of markets, competitive products (home computers), and declining "per capita" usage.
2. What is the competitive outlook? Should at least recognize the need to examine competitive dynamics. Issue areas might included: concentration of market shares; control of retail channels; and R&D capabilities (rate of new product introductions, etc.).
3. What will be the price/volume relationship in the future? Issues of prices need to be considered.

BETTER/OUTSTANDING ANSWERS

No bounds on creativity, but better answers would address:

Market Potential

- Recognize that there is a relationship between market penetration and growth in new users which, when combined, yields an industry volume estimate.
- Address the shifting mix of product purchases, in this case from hardware (player unit) to software (video cassettes).

- Seek to look at buyer behavior in key buyer segments, i.e., "fad" potential of product.

Software

- Recognize technology standards are set by industry leaders. In this situation, the division as a secondary player will have to follow these standards.
- Recognize that different distribution needs may exist for different products (In this case, hardware versus software).
- Discuss the effect capacity additions can have on overall industry price/volume relationships and on industry price levels.

Company's Ability to Compete

- Should ask what the capacity expansion is designed to do.
- Explore the cost position of the client division relative to that of other competitors.

Seek to understand reason for poor profit performance of division